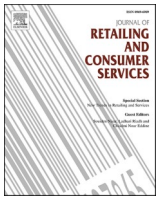
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Beyond sales: Seller explanation, product availability, and returns in livestream commerce

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ABSTRACT

Livestream commerce combines seller explanation with continued product availability in a shopping cart. Our data separately record seller explanation duration and cart availability duration; these measures do not isolate informational content or establish actual consumer exposure. Sales alone may miss differences in the post-purchase outcomes associated with these durations. We therefore examine product returns as an indicator of postpurchase outcomes, estimating associations with streamer–product and broadcast fixed effects on a panel of nearly 190,000 livestream product listings. The two durations diverge sharply: longer explanation is associated with a lower return rate and longer cart availability with a higher one, even though both are associated with higher sales. The divergence widens with the scope for uncertainty about whether a product suits a particular buyer, while the price–discount pattern does not match the specific impulse-purchasing prediction examined here. The pattern is consistent with a buyer-screening interpretation, but the data do not observe purchasers' exposure to explanation or distinguish screening from alternatives such as telepresence. Since only one of the two is scarce, the margin a seller controls is the allocation of the host's attention rather than the length of a product's stay in the cart.

1. Introduction

Open the shopping cart of a livestream broadcast and several dozen products are sitting in it at once. At that moment the host is talking about one of them. He talks for a few minutes and then moves on to the next item, and the product does not leave with his attention: it stays in the cart and remains available for purchase to viewers who find it in the cart, at no further cost in airtime. A product that has been presented therefore receives a short burst of talk followed by a long tail of availability, and most products in a broadcast receive only the tail.

The two are not the same kind of thing. Seller explanation may combine information with persuasion, endorsement, and interaction; cart availability records how long the product can be purchased, not whether viewers saw or attended to it. One is rival, because a minute of airtime given to one product cannot be given to another and a broadcast is only so long; the other is not, because the cart holds another item without displacing anything. The host's practice follows from the asymmetry — spend scarce minutes explaining a product while allowing it to remain available in the cart. This paper asks whether that tail is free to the seller as well as to the host.

The question calls for examining postpurchase outcomes alongside sales. Positive sales associations for both durations do not reveal whether their return-rate associations differ. Product returns provide information about postpurchase outcomes that sales volume alone does not capture. Returns may reflect product mismatch, but can also arise for other reasons, including return policies, delivery problems, or product damage. Marketing research has long treated returns as informative rather than merely costly in this sense (Anderson et al., 2009; Petersen & Kumar, 2009), and on that reading two sales with identical revenue and different return rates are not the same sale.

Two literatures provide the foundation for this question. Research on livestream commerce has established in detail what moves a viewer toward buying — the streamer's delivery and credibility, the trust and engagement that repeated viewing builds, the affordances of a real-time channel, scarcity and price cues (Lu & Chen, 2021; Wongkitrungrueng & Assarut, 2020; Yan et al., 2023; Zhai & Chen, 2023) — often examining purchasing outcomes. It has begun to ask whether more is always better, and has found that it is not: audience size and an anchor's exertion both show turning points, and a high-engagement broadcast atmosphere that lifts purchases is associated with lower postpurchase satisfaction (Fan et al., 2025; He et al., 2026; Yang et al., 2025). These findings establish that stronger purchasing outcomes need not imply better

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postpurchase outcomes. We build on this insight by comparing explanation and cart availability durations on both sales and return rates.

Research on product returns has concentrated on determinants that are fixed before the transaction: the return policy, posted before the buyer arrives (Janakiraman et al., 2016; Oghazi et al., 2018); the customer relationship, within which returns are a component of long-run value rather than a cost to be minimised (Petersen & Kumar, 2009, 2015); pre-purchase information such as reviews, which accumulates over months and is sought out by the buyer rather than supplied by the seller (Sahoo et al., 2018); and the buyer's own disposition to send things back (Zhou et al., 2026). Recent research also examines the seller's conduct during the sale: Feng et al. (2025) show that the live social interaction which distinguishes a broadcast from a rerun of the same content raises return rates, and read the result as persuasion working too well. This evidence motivates our complementary examination of seller explanation duration and cart availability duration within broadcasts. Livestreaming is one of the few markets in which the seller decides, minute by minute, which products to speak about and which to leave in the cart.

We consider two possible processes: bringing people into the pool of buyers and sorting them by fit before they enter it. Seller explanation and continued cart availability may relate differently to these processes, but the recorded durations do not separately measure reach and screening. The distinction matters because the established account of information and returns explains only half of what we find. On that account explanation may reduce uncertainty about whether a product will suit a particular buyer — the form of uncertainty Hong and Pavlou (2014) identify as a principal source of returns in online markets, and which varies with the search and experience attributes of the category in the sense of Nelson (1970, 1974) and, in this setting, Yang et al. (2023) — and better assessment of fit may reduce mismatch-related returns. Cart availability duration does not measure the information viewers receive, so it cannot be assumed to represent information-free exposure. We find a positive return-rate association for cart availability comparable in magnitude to the negative association for explanation duration; the opposite signs do not by themselves identify the processes behind them. One possible interpretation is that explanation discourages poorly matched purchases and thereby changes buyer composition; however, our duration measures do not identify which purchasers heard an explanation or exclude changes in individual information, persuasion, or telepresence. Under this interpretation, changes in the relative shares of buyer groups with different return propensities could generate the contrasting associations.

We examine the duration–outcome associations using 188,696 product listings from 12,558 livestream broadcasts by 160 Douyin influencers, drawn from a commercial vendor's record of a considerably larger panel. The setting is useable because the two components are recorded separately: the vendor timestamps when a product enters and leaves the cart, and reports as a distinct field how many minutes the host spent explaining it. The baseline specification includes session and streamer–product fixed effects. The coefficients are estimated from variation remaining after accounting for common session factors and time-invariant differences between streamer–product pairs; this does not require repeated listings of the same streamer–product pair within a single broadcast. What the fixed effects do not absorb is the streamer's judgement about a product at the moment of the broadcast: explanation time is allocated rather than assigned, and Section 5.3 reports what that costs the estimates. The setting closes one further channel by itself: in 45.8 percent of broadcasts every product is removed at a single common moment, so that shelf duration there is fixed by when a product was added rather than by any judgement about how it was selling.

Four findings follow. The two components are associated with the return rate in opposite directions: a one-log-unit difference in explanation time is associated with a rate 0.549 percentage points lower, and the same difference in shelf duration with a rate 0.448 points higher, each with a *t*-statistic above fourteen on the same listings. They run in the same direction on sales, at 0.612 and 0.209 log points, so sales alone would not reveal their contrasting associations with returns. The divergence widens monotonically with the scope for uncertainty about fit, by roughly an order of magnitude between the cheapest and the priciest quartile of merchandise — much of it between-category composition, with a flatter gradient surviving within categories on the availability side — a pattern consistent with, but not unique to, the proposed screening interpretation. The price–discount comparison does not support the specific prediction that the positive availability association concentrates in cheap, deeply discounted merchandise: this is the one cell of four in which the shelf coefficient cannot be distinguished from zero, although the comparison does not establish screening or exclude impulse purchasing among particular buyers.

The paper contributes to three literatures. To research on livestream commerce it adds a joint comparison of two selling activities within the same empirical specification: seller explanation and cart availability are separately recorded durations, associated with sales in the same direction and with return rates in opposite ones. This complements existing distinctions among livestream communication types and routes. The comparison also draws on research distinguishing product consideration from evaluation, examining how seller communication and availability relate to both purchases and returns (Chandon et al., 2009; Häubl & Trifts, 2000; Nedungadi, 1990). For research on product returns, our findings complement evidence linking live interaction to subsequent returns (Feng et al., 2025). We show that explanation duration and cart availability duration have opposite associations with return rates, despite positive sales associations for both. To research on information and product-fit uncertainty it contributes the observation that longer cart availability is associated with higher return rates, together with an account of why: the two durations may be associated with differences in buyer composition, although neither directly measures what purchasers saw or knew. Under the assumptions of the screening illustration, greater fit uncertainty predicts a wider divergence and negligible uncertainty predicts smaller associations; these conditional predictions organise the heterogeneity analysis without identifying screening as the underlying mechanism.

The managerial contribution is an allocation principle rather than a warning. Explanation is associated with better outcomes on both margins in every segment we examine, while additional explanation uses airtime that could be devoted to other products; shelf time is a trade-off whose terms we can put a benchmark on, and the cost of handling a return would have to be many times the unit margin before extending it stopped paying — with the threshold approximately five times as high for cheap search goods as for expensive experience goods. The principle is narrower than it first appears, because shelf duration is not a product-level lever: the cart is cleared in bulk, so what a seller controls is when a product is added and, far more consequentially, how the scarce minutes of explanation are divided across the products on the list. Section 2 develops the framework and states the hypotheses, Section 3 describes the data and the measurement of returns, Section 4 reports the results, and Section 5 the robustness checks.

2. Literature and framework

Research on retail display and consumer consideration distinguishes whether a product attracts attention or enters a consumer's choice set from how the consumer evaluates it (Chandon et al., 2009; Nedungadi, 1990). Studies of shelf facings show that the number and position of product displays can affect visual attention and evaluation, but greater attention does not necessarily translate into more favourable evaluation (Chandon et al., 2009). Eye-tracking research on print advertisements further distinguishes the capture of attention from its transfer among brand, pictorial, and text elements, indicating that attention depends on how a display is composed rather than simply on its presence (Pieters & Wedel, 2004). In memory-based choice, brand accessibility can influence consideration and choice without changing brand evaluations

(Nedungadi, 1990). Consideration-set models add an explicit cost–benefit account: consumers weigh the expected gain from considering another brand against the cost of evaluating it (Roberts & Lattin, 1991). These distinctions help situate cart availability: a longer listing creates a longer opportunity to encounter and purchase a product, but does not establish that viewers noticed or considered it.

A related literature examines how information and decision support shape consumers' evaluation of available alternatives. Research on interactive home shopping distinguishes access to a larger range of products from the ability to narrow that range and obtain useful information about the remaining alternatives; the benefit of greater selection depends partly on these evaluation capabilities (Alba et al., 1997). In online shopping, recommendation agents and comparison tools can change search effort, consideration sets, and decision quality (Häubl & Trifts, 2000). An online wine-shopping experiment further shows that making quality information easier to use can improve consumers' liking for the wines they subsequently taste, linking the shopping information environment to the quality of product choice (Lynch & Ariely, 2000). Work on product-fit uncertainty and online reviews also connects the information available before purchase to product evaluation and subsequent returns (Hong & Pavlou, 2014; Sahoo et al., 2018). This literature helps motivate examining seller explanation alongside availability. Explanation may support evaluation, but its duration also encompasses persuasion, endorsement, and interaction; it therefore does not isolate information provision or measure what individual buyers learn.

We examine seller explanation duration and cart availability duration in relation to both sales and return rates. The question is whether these selling activities exhibit different associations with postpurchase outcomes even when both are positively associated with sales. Livestream commerce provides a setting in which the two durations are separately recorded and can be compared within the same empirical specification. Attention, consideration, and information acquisition may help explain the observed associations, but the listing-level data do not directly measure these processes.

2.1. Selling activity in livestream commerce and the quantity margin

Livestream commerce has been studied primarily as a problem of generating purchases, and the mechanisms identified are those that move a viewer toward buying. One body of work concerns the streamer: the emotional contagion an online celebrity transmits (Meng et al., 2021), voice characteristics (Luo et al., 2025), linguistic emotional valence (Ma et al., 2024), narrative style (Zhang et al., 2025), and nonverbal conduct, whether measured by mining the video itself (YChen et al., 2025 Y. Chen et al., 2025), from the interaction between two hosts (Liu & Zhao, 2024), or from the anchor's body movement (Yang et al., 2025). A second concerns the format: the affordances that distinguish a broadcast from a product page (Yan et al., 2023), the relational bonds and trust that repeated viewing builds (Wongkitrungrueng & Assarut, 2020; Zhai & Chen, 2023), the engagement that follows and the customers it acquires (Zheng et al., 2022), the motivations viewers bring to watching (Zheng et al., 2023), the mental imagery a live demonstration supports for hedonic goods (Wang et al., 2022), and the immersion different broadcast genres produce (Tian & Frank, 2024). A third, closest to the mechanism we study, concerns information: livestreaming reduces consumers' uncertainty about the product (Lu & Chen, 2021), and it does so to the extent that the signals a streamer emits are mutually consistent (Chen et al., 2023).

One strand of this body of work examines purchasing and engagement outcomes, including purchase intention, engagement, conversion, units sold, sales performance, and gross merchandise value. Related studies examine scheduling, selection, and channel decisions relevant to sales: which showcasing mode a seller should adopt (Xin et al., 2023), which products belong on the running list (Su et al., 2026), and how a broadcast spills over into the seller's other channels (Wang & Chen, 2026). These studies provide a basis for examining selling decisions, while the work discussed below also differentiates communication and postpurchase outcomes. Assortment research also connects the products a retailer offers to opportunities for assessing fit: a model of online–offline competition shows that return costs can favor easy-to-fit products online, while the value of physical inspection can lead a competing store to stock harder-to-fit products (Y. Li et al., 2025). This makes the opportunity to evaluate products relevant to both the sales attracted and the returns generated. Our comparison concerns the time a given product remains available and is explained, rather than which products enter the assortment.

Other work, including studies in this journal, shows that stronger selling activity need not uniformly improve outcomes. The relationship between an anchor's body movement and sales rises and then turns (Yang et al., 2025); audience size raises sales performance only up to a point (He et al., 2026); and a high-engagement broadcast atmosphere, which raises purchases, is associated with lower postpurchase satisfaction (Fan et al., 2025). A parallel line documents the mechanism usually blamed for the latter: livestreaming encourages impulsive buying through engagement and deal proneness (X. Luo et al., 2024), through scarcity cues and anticipated emotion (Qu et al., 2023), and through streamer characteristics that diminish cognitive control (Jiang et al., 2026), with the tendency differing across product types (Zhang et al., 2024). The question of whether more exposure is better is therefore already live here. We extend this discussion through the specific comparison of explanation duration and cart availability duration, examining their associations with both sales and return rates.

Prior work also explicitly distinguishes forms of livestream communication. Q. Wang et al. (2025) separate central from peripheral routes to sales performance in live commerce, and L. Luo et al. (2024) separate informative from affective streamer topics and show that the two act differently on engagement. Both establish that what happens during a broadcast is not homogeneous. We build on these distinctions by examining a different comparison. The distinction we draw is between time devoted to seller explanation and time for which a product remains available in the cart, without equating either with the information received by viewers, and we examine both durations jointly in relation to sales and subsequent return rates.

2.2. Product returns as a measure of the quality of demand

Returns are the natural place to look for such an outcome, and not only because they are expensive. The return rate is informative rather than merely costly: it provides information about postpurchase outcomes that sales alone do not capture. Petersen and Kumar (2009, 2015) make this the basis for treating returns as a component of customer value rather than a cost to be minimised, and show that the profit-maximising return rate is not zero. Two sales with identical revenue and different return rates are not the same sale. A return rate is not a pure record of fit: return policy, logistics, product quality, return convenience, seller service and category norms can also affect it. The specification controls for streamer–product and session fixed effects, but does not eliminate all listing-level confounding. The return-rate associations therefore do not directly identify differences in buyer fit. Qualitative evidence on why consumers send products back points the same way: the dominant reasons concern the fit between the product and the buyer rather than defects in the product (Das & Kunja, 2024).

A substantial body of research examines how return policies, customer relationships, and pre-purchase information relate to product returns. One major strand concerns return policies. Leniency raises purchases and raises returns (Janakiraman et al., 2016), an effect Anderson et al. (2009) formalise as an option: a generous policy is worth something to a buyer precisely because it lets her buy while still uncertain. Within this journal the policy literature is extensive, covering leniency and consumer trust (Oghazi et al., 2018), perceived fairness (Pei et al., 2014), how the policy itself is framed (Rokonuzzaman et al.,

2021), its interaction with brand familiarity and product category (Jeng, 2017), how leniency should be customised across customers (Duong et al., 2025), and how return fees are read (Mayr et al., 2026). Adjacent work treats the operational side of the same decision: instant refunds (Martínez-López et al., 2022), freight strategy and selling format (Zhao et al., 2023), and return-freight insurance in livestream selling specifically (ZChen et al., 2025 Z. Chen et al., 2025). A second group concerns information reaching the buyer before purchase, where online reviews are shown to reduce subsequent returns (Sahoo et al., 2018). A third concerns the buyer rather than the seller: expectation disconfirmation and postpurchase dissonance vary systematically with when a purchase was made (Y. Wang et al., 2025), and in livestream commerce a consumer's pre-purchase return tendency predicts planned returns more strongly than unplanned ones (Zhou et al., 2026). These policy and information decisions can also interact: a model of consumer search and returns shows how search costs and return-processing costs shape whether product match is resolved before or after purchase (Jerath & Ren, 2025). When search can remain inconclusive and return policies adjust, more efficient search can even coexist with more returns, so information acquisition does not imply a uniformly negative relationship with returns (Jerath & Ren, 2025).

The seller's conduct during the sale has entered this literature only recently, and from the direction of persuasion. Feng et al. (2025) compare live broadcasts with reruns of the same recorded content on a large livestream platform, so that what distinguishes the two is the presence of live social interaction, and find that this interaction raises return rates substantially; the effect is stronger for more experienced streamers and for lower-priced merchandise. Their reading is that livestreaming persuades too well. Findings from adjacent settings are consistent with it: impulsive buying raises returns in online shopping generally (Li et al., 2026), and in short-video live commerce impulsive buying concentrates in identifiable product types (Zhang et al., 2024). Alongside this seller-side account, Zhou et al. (2026) supply the buyer-side complement, showing which consumers arrive already intending to return. The two together describe the demand and supply sides of the same outcome.

This emerging account establishes that selling activities can have consequences beyond the initial purchase. We build on it through a specific comparison of two recorded durations. A livestream combines seller explanation with cart availability, and our data record their durations separately. Explanation can include information as well as persuasion, endorsement, and interaction, while cart availability need not translate into actual viewing or attention. The two measures do not isolate persuasive intensity or informational content. Nevertheless, a screening account motivates examining whether their return-rate associations differ. Our contribution is to examine this contrast in the same empirical specification, using return rates alongside sales.

2.3. Seller explanation and cart availability

One possible account links explanation to product-fit assessment: communicating product attributes may reduce uncertainty about suitability and thereby reduce returns due to poor fit. Nelson (1970, 1974) supply the underlying distinction between attributes that can be evaluated before purchase and attributes that cannot; Hong and Pavlou (2014) identify product-fit uncertainty as a principal source of returns in online markets and show that it varies systematically across product types; Huang et al. (2009) show that the distinction predicts observable online behaviour, and Yang et al. (2023) show that it organises outcomes in the setting we study, since streamer attributes affect sales differently for search and for experience products on Douyin. Within livestreaming itself, Lu and Chen (2021) show that the format reduces perceived uncertainty and that this raises purchase intention, and Chen et al. (2023) that the reduction depends on the consistency of the signals a streamer emits. These are the closest antecedents to what follows, and their outcome is again a quantity: they establish that uncertainty falls, not what happens to the demand that uncertainty was holding back.

That account offers one interpretation of the explanation result. It does not yield an unambiguous prediction for cart availability: a listing may provide information, viewers may have heard an earlier explanation, and availability duration does not reveal either process. We observe a positive association between cart availability duration and the return rate, comparable in magnitude to the negative association with explanation duration. The contrasting associations motivate considering how the composition of purchasers may vary, without attributing them to a measured information contrast.

We distinguish two possible processes that could contribute to the duration–outcome associations. Reach refers to attracting potential buyers, whereas screening refers to selection into purchase on the basis of product fit; neither process is directly observed in our data. These processes need not correspond one-to-one to our measures. Explanation may attract attention as well as convey information; an available cart listing may serve viewers who have or have not heard the explanation. The vendor records cart entry and removal timestamps and a separate total explanation duration, not the timing of explanation within the availability window.

A hypothetical mixture illustrates how differences in buyer composition could generate the observed pattern. For illustration, consider purchasers who have and have not heard an explanation; these groups are not observed in our data. Those who buy without hearing the pitch may rely on other information and return at some rate r_u. Suppose that hearing an explanation provides a signal about fit, discourages some poorly matched purchases, and leaves a group of purchasers with a lower return propensity r_s; this ordering is an assumption of the illustration, not an observed difference between buyers. In this illustration, the listing-level return rate is a weighted average of the two groups' return propensities. If longer availability increases the share of purchasers who have not heard the explanation, the mixture shifts toward r_u. If longer explanation increases the share purchasing after such an assessment, it shifts toward r_s; a longer pitch may also sharpen the signal and lower r_s itself. These are interpretive assumptions, not implications of the duration definitions: the intervals may overlap, and cart purchases can follow exposure to an explanation. If each activity also attracts additional purchasers, this account can accommodate positive sales associations for both durations. Under these assumptions, changes in buyer composition could generate the opposite return-rate associations; the illustration does not establish that the proposed groups exist as described or that their shares change with either duration. We should be explicit about the standing of this account. The mixture is a claim about buyers, and the panel identifies no buyer: it records, for each listing, how many units sold and how many came back, but not whether any particular purchaser heard the pitch, nor where within the shelf window the explained minutes fell. We examine whether listing-level patterns — the opposite signs, variation with fit uncertainty, and differences across price and discount — are consistent with this interpretation, rather than treating them as a direct test of the screening process. Section 4 reports the extent of this consistency and its limits; similar patterns could arise through other processes, so these comparisons do not identify screening.

Telepresence provides another possible explanation that our design cannot separate from buyer screening. A purchaser who has not heard the pitch may have experienced different demonstrations, interaction, or sensory presentation; cart availability duration does not establish what that purchaser encountered. Research on telepresence in live-stream shopping finds that the sense of being present with a product shapes postpurchase evaluation, with higher telepresence narrowing the gap between anticipated and received utility (Barta et al., 2023), and that the immersive states the format generates bear on purchase behaviour in their own right (Dong et al., 2023). Differences in telepresence and product presentation could contribute to the observed return-rate associations alongside or independently of buyer screening; cart purchases do not establish that a buyer experienced lower telepresence. These interpretations cannot be distinguished with the available data.

Distinguishing them would require evidence on individual purchasers' exposure and evaluation processes, together with a design capable of separating their contributions (Section 6.3). We therefore retain screening and telepresence as possible interpretations without ranking their contributions; neither duration coefficient isolates either process.

H1a. Longer seller explanation duration is associated with a lower return rate.

H1b. Longer cart availability duration is associated with a higher return rate.

H1c. Both are associated with higher sales.

H1c is not a control result. It establishes the sales comparison: both durations may be positively associated with sales even when their associations with return rates differ. Repeated listings capture additional purchase opportunities, although their count does not establish consumers' exposure or the information they acquired.

One scope condition attaches to H1a and belongs where the hypothesis is stated. Explanation time is observed only for products the host chose to explain, so the hypothesis and the test concern the intensive margin — how the return rate varies with the length of a pitch among products that received one — and not the extensive margin of being explained at all. The latter is a selection the streamer makes, and products selected for a pitch differ from those left in the cart in ways that a comparison across the two would not separate from the pitch itself. H1b carries no such restriction: shelf duration is recorded for every listing, and Section 4.1.1 reports the availability estimate on the full panel as well as on the explained subsample.

The screening illustration also motivates conditional predictions about product-fit uncertainty. If an explanation changes purchase selection mainly by helping buyers assess fit, the assumed gap between r_u and r_s should be smaller where buyers can already establish fit before purchase. Under the additional assumption that the groups otherwise have similar return propensities, their rates would converge and changes in their relative shares would have little effect on the listing-level return rate. Where fit is harder to assess, selection based on an explanation could produce a larger gap. These assumptions motivate examining whether the two duration associations diverge as fit uncertainty increases and approach zero where it is negligible; such a pattern would be consistent with screening but would not uniquely identify it.

H2. As product-fit uncertainty rises, the explanation association becomes more negative and the availability association more positive; where fit uncertainty is negligible, both approach zero.

H2 provides a joint pattern against which to assess the screening interpretation, but similar patterns could also arise from unobserved differences across products or selling activities. We report in Section 4.2 that the first part of H2 holds across two differently constructed proxies, and that the second part — the approach to zero — is supported by the price proxy, where the coefficients in the cheapest decile of listings are −0.088 and 0.168 against −0.915 and 1.532 in the priciest, but only partially by the category proxy, where the compression is weaker and both coefficients remain distinguishable from zero at the bottom of the distribution.

A competing account of the availability result should be stated before it is tested, because it is the one a reader will supply. On an impulse-purchasing account, a product sitting in a cart is bought quickly on the basis of a price and a thumbnail, in a hot state that a cold state later regrets. This is the mechanism documented in this channel by X. Luo et al. (2024), Qu et al. (2023) and Jiang et al. (2026), and connected to returns by Li et al. (2026) and, for livestreaming specifically, by Feng et al. (2025). It also predicts that shelf time raises returns, but it locates the damage somewhere else. Impulse purchasing is characteristically a low-ticket, deep-discount phenomenon, and the evidence that it varies by product type points the same way (Zhang et al., 2024); screening is worth most where merchandise is expensive and hard to judge. The two accounts therefore disagree about where the harm should concentrate, and the disagreement is testable.

H3. If the availability association reflects impulse purchasing rather than unscreened demand, it is strongest for low-priced, deeply discounted merchandise. If it reflects unscreened demand, it is strongest for expensive merchandise whose fit is hard to judge.

One operational asymmetry between the two duration measures also bears on what a seller can do with these findings. Explanation is rival: a minute spent on one product cannot be spent on another, so the total is fixed by the length of the broadcast and the question is how to divide it. Shelf time is non-rival: dozens of products occupy the cart simultaneously without displacing one another, so there is no internal constraint on adding more. This is why sellers behave as they do — continued availability does not require continuous host explanation — and it is why the allocation problem in Section 4.5 is a problem about the pitch rather than about the shelf.

The components of this argument are not new. Information reduces uncertainty and uncertainty produces returns; acquisition channels differ in the quality of the customers they deliver; advertising has long been divided into what informs and what persuades. We apply these ideas to the joint examination of explanation and availability durations within a single channel. Studies separating content and communication routes within the pitch (L. Luo et al., 2024; Q. Wang et al., 2025), together with research on live interaction and returns (Feng et al., 2025), provide the foundation. Our specific contribution is to compare the two durations on sales and returns and examine how these associations vary with proxies for fit uncertainty. The setting is what makes the assembly possible: livestreaming is a market in which explanation duration and cart availability duration are recorded separately for the same product listing, although they can overlap and do not separately identify reach and screening.

3. Data

3.1. Setting, panel and the measurement of returns

Livestream commerce on Douyin, the Chinese platform operated by ByteDance, works differently from the retail environments in which product returns have mostly been studied. A host broadcasts for several hours at a time and sells from a pinned cart that viewers open in a side panel. Products enter the cart one at a time, at a moment the host chooses, and they mostly leave it together: 86.6 percent of listings are removed at their broadcast's final removal moment, and the median broadcast has two distinct removal moments for a median of 38 products. The cart is therefore cumulative — a median of 31 products sit in it simultaneously — and while a product sits there any viewer can buy it without the host ever mentioning it. Separately, and for only a fraction of the products, the host takes the item in hand and talks about it — describing the material, trying it on, answering questions typed into the comment stream — and then moves on to the next item. The product does not leave with the host's attention. It stays in the cart and remains available for purchase to viewers who find it in the cart, at no further cost in airtime. A pitched product therefore receives a short burst of talk followed by a long tail of availability: the median explanation runs 5 min against a median shelf stay of 136, or 4.5 percent of the window. Sixty-nine percent of listings receive only the tail. The two are not the same thing, the vendor records them separately, and the distinction is what this paper is about.

Our data come from a commercial vendor that tracks this market at the level of the individual product listing, which is also our unit of analysis: one occasion on which one streamer places one product in the cart during one broadcast. The extract covers 164 macro-influencers, each with more than ten million followers and between 10.1 and 98.5 million, spanning the content genres through which selling on this platform is organised — short-form drama, food, film and

entertainment, lifestyle and education — over the period from 25 April 2022 to 11 November 2022. These are among the platform's largest sellers, so the panel is not drawn from the tail of the market; it is equally not a random sample of sellers, and we return to what that means for external validity in the concluding section. The panel contains 617,440 listings across 12,936 broadcasts, 93,662 distinct products and 3463 product categories, with a median of 38 listings per broadcast and 52 broadcasts per streamer.

Two features of this structure supply the comparisons the estimator uses. The same product is listed repeatedly by the same streamer: 68.1 percent of the 99,352 streamer–product pairs are aired at least twice. In the estimation sample of Section 4, which is smaller, 45.3 percent of pairs are aired at least twice and 79.8 percent of listings sit in a pair aired more than once, the remaining fifth being singletons that the streamer–product effect absorbs entirely. And a single broadcast carries many products treated very differently within it — some explained at length, some never mentioned, some added early and some late — so a session effect can be estimated without absorbing the variation of interest. Cart availability duration, the interval between the listing and delisting timestamps, is available for every listing; seller explanation duration is recorded for 30.6 percent of them, with only 2.9 percent of broadcasts recording none at all. Section 4.1 sets out what that coverage is and is not and reports the estimates under each way of handling it. One further feature bears on how much of shelf duration is discretionary: in 5929 broadcasts, 45.8 percent of the total, every product is removed at a single common moment when the host closes the broadcast and clears the shelf, so that shelf duration is fixed by when a product was added rather than by any judgement about how it was selling. Section 4.3 uses this.

The dependent variable is the vendor-reported return rate attached to each listing, expressed in percentage points. We interpret it as the share of sold units subsequently returned, subject to the attribution limitation discussed below. Two properties can be checked against the data. It is recorded exactly rather than in bands — the vendor reports units sold in bounded intervals such as “fewer than 10” or “200+”, but gives the return rate to two decimal places for every listing, so the variable the paper explains is not interval-censored. Variation across successive listings does not establish whether the reported rate is calculated cumulatively: a cumulative ratio can decline as its denominator changes. We therefore do not use monotonicity to establish the rate’s attribution to individual listings. The reported rates also agree in ordering with an independently recorded figure — the vendor also reports a return rate for each broadcast as a whole, available for 12,641 of the 12,936 sessions, and aggregating our listing records to the broadcast reproduces its ordering with a rank correlation of 0.88; the same aggregation on units sold reproduces the broadcast total with a rank correlation of 0.99 and a median ratio of 1.01, so the listing records account for essentially all of a broadcast's activity.

The two return figures agree in ordering but not in level, and the gap comes from the numerator. Counts are banded, and the bands fall differently on the two quantities: 27.2 percent of listings report returned units as fewer than ten against 21.9 percent for units sold, while units sold are far more often in an open-ended upper band. Substituting a midpoint of five for a true value of one or two is a large proportional error and it falls on returns precisely where returns are small, which is most of the time. We therefore build nothing on aggregated counts: the analysis uses the listing-level rate, which is recorded exactly. Two limitations cannot be resolved from inside the data at all. The first concerns the pitch: the vendor records how many minutes a product was explained but not when within the shelf window those minutes fell, so we cannot separate the shelf time that precedes a pitch from the shelf time that follows it, nor identify which cart purchases were made by viewers who had already heard one. The second concerns the outcome. The vendor does not disclose the window over which a return is attributed to a listing; if that window is short relative to the interval between a pair's airings the measure is close to the object we want, and if it is long some of what is attributed to a listing was decided later. Nothing in the panel identifies the window, and we treat this as a limitation rather than something we can bound.

3.2. Variables and descriptive statistics

Every variable is obtained by a mechanical rule from a field the vendor records; none involves a judgement of ours about what a product is. Table A1 in the appendix gives the definitions and the coverage of each, setting out in full the three that are constructions rather than raw fields: the airing number, the category baseline return rate and discount depth.

Seller explanation duration records time spent explaining or presenting the product; it may combine information, persuasion, endorsement, demonstration, and interaction. Cart availability duration records the interval between addition to and removal from the cart, not realised consumer exposure. We do not observe whether viewers opened the cart, whether the product was visible without scrolling, or how much attention it received, and cannot weight the interval by the viewers present throughout it. The two durations may overlap, and the data do not reveal when the explanation occurred within the availability window or which purchasers heard it. Cart listings may themselves convey information. We therefore do not interpret these measures as informational versus noninformational exposure.

Table 1 describes the estimation sample of Section 4 alongside the full panel, because Section 4 restricts attention to listings with a recorded explanation duration and the reader should be able to see what that restriction does. The restricted listings are the ones the host actually presented: a median of 200 units sold against 10 across the panel, 136 min on the shelf against 167, three airings in their own history against seven, and a mean return rate of 3.43 percent against 2.94.

Two things the data cannot support should be said here rather than left to the reader to discover. The panel carries no order-level or buyer-level identifiers, so a return is observed as a rate attached to a listing and never as an event attached to a person: we cannot distinguish one buyer returning three items from three buyers returning one each, cannot follow a buyer across purchases, and cannot observe repurchase. That rules out the natural extension of this paper — whether cart availability duration damages customer value beyond the transaction it generates — and we say so in the concluding section rather than approximate it with something the data do not contain. And the return rate recorded for a substantial group of streamers sits close to zero throughout: 74 of the 160 streamers in the estimation sample have a mean recorded return rate at or below one tenth of one percent, and they account for 39 percent of its listings. That group is concentrated among the largest accounts but is not confined to them. We document it and re-estimate without it in Section 4.4 rather than treat it as a behavioural fact.

4. Empirical results

4.1. Seller explanation duration and cart availability duration

Our data record how long the host explains or presents a product and how long it remains available in the shopping cart. These durations may overlap and do not measure what individual viewers saw or attended to. To characterise how each is associated with the quality of the demand a listing captures, we estimate

$$y_{ips} = \beta_1 \,\text{explanation}_{ips} + \beta_2 \,\text{availability}_{ips} + \beta_3 \,\text{repeat}_{ips} + X'_{ips}\gamma + \alpha_{ip} + \delta_s + \varepsilon_{ips} \quad (1)$$

where y is the return rate of the listing of product p by streamer i in session s, explanation is log(1 + explanation minutes), availability is log (1 + cart availability minutes), and repeat is the log of the airing number, the ordinal index of the current airing within each streamer–product pair’s observed history, including the current airing. X collects

Table 1
Summary statistics.

	M	*SD*	p25	Median	p75	*N*	Full-panel mean
Panel A. Outcomes and durations							
Return rate (pp)	3.43	8.15	0.00	0.06	2.67	188,698	2.94
Explanation time (min)	17.87	124.73	2.27	5.02	10.65	188,698	17.87
Shelf duration (min)	287.50	597.68	48.85	136.37	299.97	188,698	367.07
Airing number	12.95	32.66	1	3	9	188,698	34.65
Sales (units)	2157	26,281	30	200	1000	188,698	755
Panel B. Controls							
Price (RMB)	1433	107,980	19.90	59.90	139.90	188,698	2644
Discount depth	0.05	0.15	0.00	0.00	0.00	188,698	0.04
Branded listing (0/1)	0.76	0.43	1	1	1	188,698	0.73
Relative position	0.36	0.27	0.12	0.29	0.54	188,696	0.51
Category baseline return (pp)	3.07	2.88	1.09	2.13	4.17	183,872	3.06
Product age (days)	70.11	128.63	0.15	8.70	74.07	188,698	128.80
Commission rate (%)	2.75	7.27	0.01	0.20	1.00	150,762	4.33

Panel C. Panel structure	Estimation sample	Full panel
Listings	188,698	617,440
Commerce sessions	12,558	12,936
Streamers	160	164
Distinct products	67,117	93,662
Streamer–product pairs	69,552	99,352
Product categories	2971	3463

Notes: The estimation sample is the set of listings for which the vendor records an explanation duration; the full panel is every listing. Variables are as defined in Table A1, before the logarithmic transformations. Price and sales are extremely right-skewed, which is why both enter the estimating equation in logs; the price mean is inflated by a small number of very high-value listings and the median is the more informative figure. Units sold are reported in bounded intervals for 72.4 percent of listings: open-ended bins such as "200+" are entered at their lower bound and the "fewer than 10" bin at its midpoint of five, so the reported dispersion understates the true dispersion. In the estimation sample the median session carries nine listings and the median streamer–product pair is observed once; 29.4 percent of its listings record no returns at all, against 57.6 percent across the full panel.

log price, discount depth, a branded-listing indicator and the listing's relative position in the running order. α_{ip} is a streamer–product fixed effect and δ_s a session fixed effect. Errors are clustered at the session level. All duration variables enter as log(1 + minutes) rather than log (minutes) so that slots of arbitrarily short duration remain in the sample; every interdecile range quoted below is computed on that transformation. Table A1 in the appendix gives the definition of each variable and the share of listings for which it is recorded.

Equation (1) is estimated on 188,696 of the 188,698 listings for which the vendor records an explanation duration — 30.6 percent of the 617,440 listings in the panel, spanning 12,558 sessions and 160 streamers; two listings are dropped for a missing control. The restriction requires justification, because the remaining listings are not missing at random. Listings with no recorded explanation duration have a median sales volume of five units against two hundred for those with one, appear a median of twelve times in their pair's history against three, and sit on the shelf longer. They are the repeatedly re-listed catalogue tail: products placed in the cart and never pitched. The explanation coefficient is accordingly an intensive-margin estimate, identified from variation in the length of a pitch among products that received one; it does not speak to the difference between being explained and not being explained, which Section 2.3 sets outside the paper's scope.

It is tempting to retain them by entering an indicator for whether a product was pitched at all alongside the log duration set to zero elsewhere, and this raises the nominal sample to 616,216. We retain the restricted sample as the baseline because the research question concerns variation in recorded explanation duration. Setting unrecorded durations to zero and adding a pitched indicator combines this intensive-margin comparison with differences in whether explanation is recorded. Although the duration regressor is constant among the 427,520 added listings, these observations can still affect its coefficient through the estimation of controls and fixed effects. Moreover, the pitched indicator reflects the streamer's selection decision and is not an exogenous treatment. The paper's claim is that two duration measures carry opposite signs, and a claim about two coefficients is most credible when both are estimated from the same equation on the same observations. Section 4.1.1 reports every handling side by side and compares the two specifications in which seller explanation duration actually appears; the availability coefficient changes from 0.448 to 0.493.

A word on what equation (1) does and does not deliver. The coefficients are within-streamer–product, within-session associations between its explanation and availability durations and how its demand behaved. They are not causal effects, and we do not present them as such. The fixed effects rule out the two most obvious confounds — that products given long pitches are simply different products, and that sessions with long pitches are simply busier sessions — and Section 4.3 closes one further channel by restricting attention to sessions in which the delisting moment is not chosen. What remains is an association that survives a demanding set of controls, together with a pattern across subsamples in Section 4.2 that a confound would have to reproduce in two opposite directions at once. That is the basis on which we ask the reader to take the interpretation seriously, and it is weaker than identification.

Two features of the specification deserve emphasis. First, the two fixed effects do different work and both are necessary: the streamer–product effect holds the merchandise and the seller constant, so the estimates cannot reflect the possibility that products receiving long pitches are simply better products, while the session effect holds constant the traffic, promotional intensity and audience composition of the broadcast. Because a single session contains products at many different points in their own histories and receiving very different amounts of airtime, the two effects do not absorb each other. Second, the two durations must enter together. They are correlated at 0.168 after the fixed effects are removed, and because they carry opposite signs on the return rate, omitting one biases the other toward zero. Columns (1) and (2) of Table 2 show this directly: the availability coefficient is 0.319 when seller explanation duration is left out and 0.448 when it is included, an attenuation of nearly thirty percent.

Table 2 shows two things at once. On the return rate the two durations point in opposite directions: a one-log-unit difference in seller explanation duration is associated with a rate 0.549 percentage points lower, the same difference in cart availability duration with a rate 0.448 points higher, with *t*-statistics of −15.8 and 14.3 on the same 188,696 listings across 12,558 sessions. On sales they do not: both are positively

Table 2
Seller explanation and cart availability — return rate and sales.

Panel A. Return rate (pp)	(1)	(2)	(3)
log(explanation duration)		−0.549***	−0.441***
		(0.035)	(0.035)
log(cart availability)	0.319***	0.448***	0.396***
	(0.030)	(0.031)	(0.035)
log(airing number)	0.343***	0.231***	0.181***
	(0.040)	(0.040)	(0.051)
log(price)	0.131**	0.125**	0.265***
	(0.057)	(0.057)	(0.081)
Discount depth	−0.172	−0.117	0.114
	(0.134)	(0.133)	(0.149)
Branded listing	−0.100	−0.099	−0.278
	(0.291)	(0.291)	(0.407)
Relative position	0.659***	0.270***	0.330***
	(0.081)	(0.083)	(0.092)
Commission rate			0.007
			(0.026)
log(product age)			0.037
			(0.036)
R^2	0.726	0.727	0.717
Observations	188,696	188,696	150,760
Sessions	12,558	12,558	11,877
Panel B. Log sales	**(1)**	**(2)**	**(3)**
log(explanation duration)		0.612***	0.602***
		(0.005)	(0.005)
log(cart availability)	0.353***	0.209***	0.218***
	(0.006)	(0.005)	(0.006)
log(airing number)	−0.277***	−0.151***	−0.150***
	(0.007)	(0.006)	(0.007)
log(price)	−0.225***	−0.218***	−0.441***
	(0.019)	(0.018)	(0.034)
Discount depth	0.206***	0.144***	0.102***
	(0.025)	(0.022)	(0.026)
Branded listing	0.119*	0.118*	0.099
	(0.069)	(0.063)	(0.073)
Relative position	−0.512***	−0.078***	−0.119***
	(0.017)	(0.015)	(0.015)
Commission rate			−0.000
			(0.003)
log(product age)			−0.015***
			(0.005)
R^2	0.936	0.951	0.954
Observations	188,696	188,696	150,760
Sessions	12,558	12,558	11,877

Notes: Cluster-robust standard errors (session level) in parentheses. ***$p < .01$, **$p < .05$, *$p < .10$. All columns include session and streamer–product fixed effects and are estimated on listings with a recorded explanation duration, two of which are dropped for a missing control. Column (2) is the preferred specification. Column (3) adds the commission rate and product age, recorded for 79.9 percent of the sample; re-estimating column (2) on that restricted sample without the two controls gives −0.442 and 0.395, so the difference between columns (2) and (3) is one of sample rather than of controls. The return rate is winsorised at the 99th percentile. R^2 is that of the full model including the fixed effects. The largest variance inflation factor in the design is 1.13 and the condition number is 1.46. Coefficients are obtained by residualising on the fixed-effect dummies with sparse LSMR rather than by iterated demeaning, which converges too slowly at this design's dimensions.

associated with it, at 0.612 and 0.209 log points. That divergence is the paper's central observation. A manager watching the sales line alone would see one margin moving in the same direction for both durations; the return rate reveals a second margin on which they separate, although neither outcome identifies whether purchasers heard an explanation or attended to the cart listing.

Repetition patterns with availability rather than seller explanation duration. The airing number enters in logs, so doubling the cumulative airing number, which includes the current airing, is associated with a return rate 0.16 points higher and sales 0.10 log points lower, each further airing contributing progressively less. The sign pattern places repeated airings on the side of availability rather than persuasion, which is what one would expect if what accumulates across airings is reach into progressively less well-matched buyers.

The magnitudes are substantial on both sides. The interdecile range of seller explanation duration is 2.577 log units, from 0.9 to 24.4 min, and that of cart availability duration is 4.471 log units, from 8.4 to 824.2 min. Across the first, the listing with the longer pitch has a return rate 1.41 percentage points lower, or 41 percent of the subsample mean of 3.43 percent; across the second the difference is 2.00 points higher, or 58 percent of the mean. The airing number accounts for a further 0.78 points, or 23 percent, across a range running from a product's first airing to its thirtieth. Computing these figures from log(minutes) rather than log(1 + minutes) would overstate the explanation range by roughly a quarter and leave the availability range almost unchanged.

4.1.1. *Alternative handling of unrecorded explanation time*

Because the sample restriction is the most consequential design decision in the paper, we report what happens under each of the ways the unrecorded listings can be handled. Row (a) is the preferred specification. Row (b) retains the full panel by entering a pitched indicator alongside the log duration, which is set to zero for unpitched listings. Row (c) drops explanation duration from the equation altogether, which is the only handling that uses the full panel without asserting anything about what an unrecorded duration means. Row (d) applies that same agnostic handling within the pitched subsample, isolating the change of sample from the change of specification.

Three points follow. The comparison that bears on the sample restriction is between rows (a) and (b), because those are the two in which seller explanation duration appears in the equation at all. The availability coefficient is 0.448 on the pitched subsample and 0.493 on the full panel, a difference of 0.045. The coefficient remains positive and statistically significant in both specifications.

The explanation coefficient is −0.549 on the subsample and −0.479 on the full panel with the indicator, a difference of 0.070. The coefficient remains negative and statistically significant in both specifications. We prefer the former and note the latter. The full-panel figure is a weighted combination in which nearly seventy percent of the observations contribute no variation to the coefficient but do contribute to the fixed-effect projections and to the estimated variance, and it sits alongside an extensive-margin term whose positive sign — 0.410 on the return rate — reflects the streamer's decision about which products to promote rather than any effect of promotion itself.

Rows (c) and (d) exist to forestall a misreading of the first two, and the arithmetic is worth setting out. Within the same 188,696 listings, dropping seller explanation duration from the equation moves the availability coefficient from 0.448 to 0.319: the two durations correlate at 0.168 within streamer–product and session and carry opposite signs, so a specification containing only one of them recovers an attenuated version of it. On the full panel the same omission moves the coefficient only from 0.493 to 0.453, a smaller specification difference that does not by itself identify the reason for the change. Rows (a) and (c) therefore sit at 0.448 and 0.453 not because the restriction is innocuous but because two differences — one of sample and one of specification — happen to offset. We state this explicitly because those two figures would otherwise be the natural pair to quote, and quoting them as a robustness check would be wrong.

4.2. *What governs the gap: product-fit uncertainty*

If the divergence in Table 2 arises partly through differences in buyers' assessment of fit associated with explanation and availability, the size of the divergence should depend on how much uncertainty there is to resolve. Under the assumptions in Section 2.3, higher fit uncertainty predicts a more negative explanation association and a more positive availability association; this joint prediction provides a check on the interpretation, not a unique test of screening. We use two observable proxies for the main analysis, each numeric, recorded by the vendor, and

requiring no judgement on our part: the listing's price, and the baseline return rate of its product category. Neither is a direct measure of fit uncertainty; each is a quantity that should move with it, and the paragraphs below set out what each does and does not capture.

The category measure requires a fuller account, since three parts of the paper lean on it. The underlying construct is the distinction between search goods, whose relevant attributes can be evaluated before purchase, and experience goods, whose attributes cannot (Nelson, 1970, 1974). In online retail the operative form of the distinction is product-fit uncertainty — the buyer's inability to establish before receipt whether an item will suit her in particular — which Hong and Pavlou (2014) identify as a principal source of returns in online markets and show to vary systematically across product types. Huang et al. (2009) demonstrate that the search–experience distinction predicts observable online browsing behaviour, and Anderson et al. (2009) document that realised return rates differ sharply across categories for the same reason. In livestream commerce specifically, Yang et al. (2023) show on Douyin data that the same distinction organises how streamer attributes translate into sales, which is the closest existing use of the split in this setting and the reason we adopt a category-level rather than an item-level construction.

We do not assign categories to the two classes ourselves, because any such assignment would import our own judgement into the measurement. We let realised returns do the sorting instead. For each listing we compute the mean return rate of its product category across all other streamers in the panel, excluding the focal streamer's own listings from her own value, so that the measure carries no information about her outcomes. A category in which buyers frequently discover after receipt that the item does not suit them records a high value; one in which they do not records a low value. The vendor resolves categories finely — 3463 of them, at the level of running shoes rather than footwear — so the measure is not blunted by aggregation.

Whether the resulting variable is a useable proxy for the search–experience distinction is an empirical question, and Table 4 answers it by listing the categories at each end of the distribution. The ten lowest are custom tableware, mobile top-up cards, sauces and condiments, wet wipes, puffed snacks, instant noodles, toothpaste, instant rice noodles, cookware and facial tissue — goods whose fit is known at the moment of purchase and that effectively cannot disappoint on that dimension. The ten highest are led by casual and skate shoes, running shoes, down jackets, facial-care sets, shoulder bags and knitwear — goods whose suitability turns on size, on skin type, on how an item looks on a particular person, and which cannot be established until the parcel arrives. The broad ordering is the one the theory predicts, and it was not imposed.

Two entries deserve comment rather than concealment, because they mark the limits of what the measure is. Custom tableware records a return rate of exactly zero across 923 listings; personalised goods are ordinarily exempt from the no-reason return window in this market, so its position at the floor reflects a returns policy and not a buyer's certainty about fit. Baijiu sits among the ten highest at 5.61 percent although it is a standardised branded commodity whose attributes are perfectly known before purchase; its returns plausibly arise from breakage, from authenticity disputes and from the regret attached to expensive gift purchases rather than from fit. The variable therefore measures realised return propensity, which is correlated with product-fit uncertainty but is not identical to it. Classification influenced by policy or logistics may affect the estimated gradient. We use coarse bins and compare the results with those based on price; these comparisons assess sensitivity to the proxy but do not eliminate these concerns.

Table 4
Categories at the extremes of the category baseline return rate.

Product category	Baseline return rate	Listings
Panel A. Ten lowest (search-like)		
Custom tableware	0.00%	923
Mobile top-up cards	0.12%	806
Sauces and condiments	0.70%	1256
Wet wipes	0.94%	804
Puffed snacks	0.98%	960
Instant noodles	1.33%	1111
Toothpaste	1.43%	1058
Instant rice noodles	1.53%	2287
Cookware and kettles	1.55%	1066
Facial tissue	1.73%	1543
Panel B. Ten highest (experience-like)		
Sweatshirts and fleece	4.63%	847
T-shirts	4.75%	1621
Liquid serum	5.54%	1958
Baijiu (domestic spirits)	5.61%	2292
Knitwear	6.10%	1005
Shoulder bags	6.86%	1463
Facial-care sets	7.06%	2999
Down jackets	7.88%	1209
Running shoes	9.37%	806
Casual and skate shoes	13.27%	1404

Notes: The figure displayed is the mean return rate of the category over all streamers in the panel. The regressor used throughout is the leave-focal-streamer-out version defined in Table A1, which varies by listing and therefore cannot be tabulated as a single number per category; the two differ by the focal streamer's own contribution, which is at most one streamer's share of the category. Ranking is restricted to the 37 categories with at least 800 listings in the estimation sample, so that the extremes are not occupied by categories observed a handful of times; two residual categories whose vendor labels are uninformative ("other" and "mix-and-match bundle") are omitted from the display but retained in all estimation. Category labels are translated from the vendor's Chinese taxonomy.

For the analysis we bin rather than interact. In this section the measure is cut at its terciles, 1.41 and 3.39 percent, and equation (1) is estimated separately within each bin, so that no functional form constrains how the slope varies. In Section 4.5 it is cut at its median, 2.13 percent, with listings below the cut labelled search-like and those above experience-like. Those two labels are shorthand for the two halves of this empirical distribution and are not claims about any individual category's place in a theoretical taxonomy.

Table 5 reports the results, and three patterns merit discussion. First, the gradient is present and large in all three proxies. Across price quartiles the explanation coefficient runs from −0.137 to −1.391 and the availability coefficient from 0.200 to 1.202; across the category terciles from −0.248 to −1.171 and from 0.238 to 0.808; and across

Table 3
Alternative handling of unrecorded explanation time.

Handling of unrecorded listings	log(explanation duration)	log(cart availability)	Pitched (0/1)	Obs.	Sessions
(a) Excluded [preferred]	−0.549*** (0.035)	0.448*** (0.031)	—	188,696	12,558
(b) Retained, pitched indicator	−0.479*** (0.026)	0.493*** (0.025)	0.410*** (0.057)	616,216	12,935
(c) Retained, explanation dropped	—	0.453*** (0.025)	—	616,216	12,935
(d) Excluded, explanation dropped	—	0.319*** (0.030)	—	188,696	12,558

Notes: Cluster-robust standard errors (session level) in parentheses. ***$p < .01$, **$p < .05$, *$p < .10$. Dependent variable is the winsorised return rate in percentage points. All rows include the full control set, session fixed effects and streamer–product fixed effects. In row (b) the log duration is set to zero for listings with no recorded duration; among those listings the variable has a standard deviation of zero and contributes no identifying variation to the explanation coefficient. The pitched indicator is defined only in row (b), where it also carries a coefficient of 0.445 (0.014) on log sales; it is a choice variable of the streamer and is not interpreted causally.

Table 5
Moderation by proxies for product-fit uncertainty.

	log(explanation duration)	log(cart availability)	Observations
Panel A. Price quartile			
Q1 (cheapest)	−0.137*** (0.037)	0.200*** (0.042)	50,052
Q2	−0.295*** (0.060)	0.177*** (0.054)	46,945
Q3	−0.480*** (0.070)	0.407*** (0.066)	44,821
Q4 (priciest)	−1.391*** (0.106)	1.202*** (0.101)	46,878
Panel B. Category baseline return rate			
Low (search-like)	−0.248*** (0.046)	0.238*** (0.052)	61,318
Mid	−0.373*** (0.050)	0.289*** (0.045)	61,312
High (experience-like)	−1.171*** (0.086)	0.808*** (0.064)	61,240
Panel C. Discount depth			
No discount	−0.485*** (0.037)	0.400*** (0.033)	165,912
Shallow	−0.739*** (0.144)	0.482*** (0.165)	11,209
Deep	−1.398*** (0.268)	0.570*** (0.146)	11,575

Notes: Cluster-robust standard errors (session level) in parentheses. ***$p < .01$, **$p < .05$, *$p < .10$. Each row is a separate estimation of equation (1) on the indicated subsample, including the airing number, the full control set, session fixed effects and streamer–product fixed effects. Panels A and C partition the estimation sample and their observation counts sum to 188,696. Panel B sums to 183,870 because the leave-streamer-out category baseline return rate is undefined for listings in categories no other streamer in the panel carries, which is 2.6 percent of the subsample.

discount depth from −0.485 to −1.398 and from 0.400 to 0.570. The effects in the priciest quartile are roughly 10 times those in the cheapest on the explanation side and 6 times on the availability side; Section 5.4 shows that most of this spread is between-category composition, and reports what remains when price position is measured within the category. Five of the six series are monotone; the single exception is the availability coefficient between the first and second price quartiles, where 0.200 and 0.177 differ by 0.023 against standard errors of 0.042 and 0.054 and the ordering is therefore not identified. Because these are separate estimations on disjoint subsamples rather than points on an imposed line, the gradient is a property of the data rather than of the specification.

Second, and more informative than the gradient itself, the two coefficients move together and in opposite directions. With the one exception just noted, every step that makes the explanation coefficient more negative makes the availability coefficient more positive, in the same subsample, at the same time. Unobserved factors or alternative processes affecting the two associations differently could also generate this pattern. The symmetry is consistent with the proposed interpretation, although the duration measures do not establish how much a buyer knew about fit at purchase.

Third, the three proxies are not equally informative and we weight them accordingly. Price and category baseline are the two we build on, because they capture different aspects of product-fit uncertainty. Neither is exogenous: price can reflect selling decisions, and the leave-streamer-out category baseline can share category-level confounding even though it excludes the focal streamer's outcomes. Discount depth is monotone here but we treat it as corroborative rather than probative, both because the two discounted bins are small and because discount depth is itself a merchandising decision that could correlate with uncertainty through channels other than the one we propose. It appears in the table chiefly because it is the proxy on which the competing explanation in Section 4.2.1 turns.

Fig. 1 plots the subsample estimates with 95 percent confidence intervals. We visualise them because the object of interest is not the level of either coefficient but the way the two diverge as uncertainty rises, and a figure makes the symmetry of that divergence legible in a way a column of numbers does not. The solid series is the explanation coefficient and the dashed series the availability coefficient; the two fan apart across both panels, the vertical gap between them widening at every step, and neither series crosses zero. The availability series itself dips slightly between the first two price quartiles, which is the non-monotonicity noted above; the divergence between the two series does not.

4.2.1. *Examining an impulse-purchasing interpretation*

A natural competing explanation for the availability-duration result invokes impulse purchasing rather than unresolved uncertainty. On that account a product sitting in the cart is bought quickly, on the basis of a price and a thumbnail, with limited deliberation, although a host's pitch may itself contain persuasive cues; the purchase is made in a hot state and regretted in a cold one, and the return follows. The account is plausible and well established in this literature (Li et al., 2026; X. Luo et al., 2024; Qu et al., 2023), and it has direct support in this channel: Feng et al. (2025) find that the live social interaction distinguishing a broadcast from a rerun raises return rates, and that the effect is strongest for lower-priced merchandise. It makes a prediction that differs sharply from ours. Impulse purchasing is characteristically a low-ticket, deep-discount phenomenon. Uncertainty resolution is the opposite — it is expensive and hard-to-judge merchandise for which information is worth having. The two accounts therefore disagree about where the damage from cart availability duration should concentrate.

The comparison does not support the specific price–discount prediction examined here. In the cell that account identifies as its epicentre — cheap merchandise carrying a deep discount — the availability coefficient is 0.073 with a standard error of 0.142, the only one of the four that cannot be distinguished from zero and roughly one sixteenth of the 1.188 recorded for expensive merchandise carrying a deep discount. If cart availability duration damaged demand quality by inviting

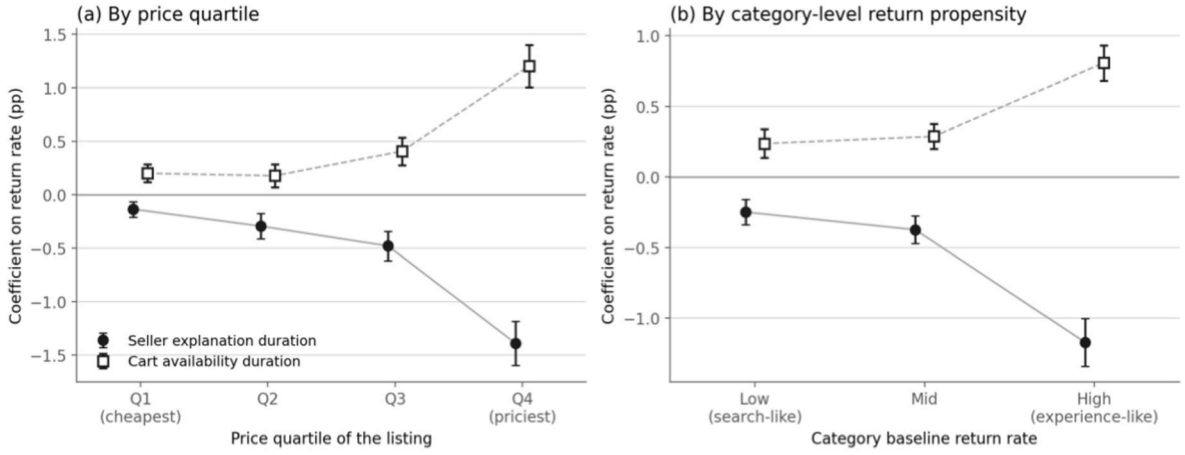


Fig. 1. The two duration measures across product types.

unconsidered purchases of cheap discounted goods, this is where the damage would be largest; instead it is where it is smallest, and where the point estimate is closest to no damage at all. The ordering across the four cells tracks price far more closely than it tracks discount, which is what an account resting on the value of resolving fit uncertainty predicts and what an account resting on hot-state purchasing does not. The comparison weighs against this specific price–discount prediction for cart availability, but it neither identifies screening nor establishes which process dominates; impulse purchasing and other processes may still contribute. The two sets of results are in fact easier to reconcile than to oppose. Feng et al. (2025) measure the intensity of live interaction, which is concentrated in the pitch and in the hot state it creates, and find the damage where an impulse account predicts it, among cheap merchandise; we measure cart availability duration jointly with explanation duration, without distinguishing purchasers by whether they heard the pitch, and find the damage where a screening account predicts it, among expensive merchandise. Both channels can operate at once; our comparison complements the evidence on live interaction rather than implying that prior research treats exposure as homogeneous.

One qualification belongs with this conclusion, and it concerns what a product-level test can see. Impulsiveness is a property of buyers rather than of merchandise: the immersive states a broadcast creates reduce deliberation more for some viewers than for others, and the flow experience that mediates impulse buying in this channel varies across individuals (Dong et al., 2023). A comparison across price and discount cells averages over whatever mixture of buyers each cell attracted, so an effect concentrated in a segment of viewers can be present and still leave a near-zero coefficient at the level we observe. The panel carries no buyer-level identifiers (Section 6.3), so we cannot partition buyers by decision style and have not attempted to. Table 6 shows that the positive availability association does not concentrate where the specified impulse account predicts; this product-level comparison does not identify the dominant mechanism or establish whether impulse purchasing affects particular buyers.

4.3. Sessions in which the delisting moment is not chosen

Of the regressors in equation (1), cart availability duration is the one most exposed to the streamer's discretion. A host who sees a product performing badly may pull it from the cart, and one who sees a product performing well may leave it up; either behaviour would confound the estimate, and neither is absorbed by fixed effects that hold constant the product, the seller and the broadcast but not the seller's within-session reaction to what is happening.

The data offer a way around this. In 5929 of the 12,936 commerce sessions in our panel — 45.8 percent — every product on the cart is removed at a single common moment, when the host closes the broadcast and clears the shelf. Within these sessions the removal moment is not a product-level decision: shelf duration is the arithmetic difference between the session's single closing time and the moment the product was added, so the channel through which discretionary delisting could operate is closed by construction. The listing moment remains the host's choice, and we return to that limitation below.

Table 7 reports an availability coefficient of 0.478 in the common-delist subsample against 0.448 in the full estimation sample, a difference well inside one standard error, and the explanation coefficient moves from −0.549 to −0.467. Restricting attention to sessions in which that one channel is closed leaves both estimates essentially where they were. This narrows the set of explanations that could produce the association; it does not establish that shelf duration causes returns, and we do not claim that it does. The standard errors roughly quintuple on the availability term because the subsample contains a fifth of the observations and, more importantly, because within a common-delist session shelf duration varies only through listing order; the coefficient remains significant at conventional levels but the test is a weaker one and we present it as such.

One limitation should be stated precisely rather than left implicit: common delisting removes discretion over the end of a product's shelf life, not over the beginning, since the host still chooses when in the running order to add each item. Shelf duration in these sessions is a deterministic function of listing time, which is why every specification controls for the listing's relative position and why we regard that control as doing substantive rather than decorative work.

4.4. Where the effects bind

The estimates so far describe an average across a panel that is far from homogeneous. Table 8 crosses the two dimensions along which we find the largest differences — the uncertainty of the product category and the size of the influencer's audience — because a single-dimension split cannot distinguish a genuine difference in behaviour from a difference in what each group happens to sell.

Two things follow. Among streamers below twenty million followers the pattern is the one Section 4.2 describes, and it strengthens with the category proxy as predicted: within the ten-to-thirteen-million tier the explanation coefficient moves from −0.348 in low-uncertainty categories to −1.517 in high-uncertainty ones, and within the thirteen-to-twenty-million tier from −0.437 to −1.738. The availability coefficient rises in step, from 0.211 to 1.240 and from 0.539 to 1.258. Three of these four series are monotone across the three category cells; the explanation coefficient in the smallest tier dips to −0.258 in the middle cell before falling to −1.517 in the highest, so the endpoints order as predicted while the intermediate step does not. The contrasting associations also appear within these streamer-size groups.

Among streamers above twenty million followers the coefficients collapse in every category cell — by roughly an order of magnitude in most of them and by two orders for the explanation coefficient in high-uncertainty categories, where it is −0.015 against −1.517 and −1.738 in the other two tiers. We initially read this as a behavioural scope

Table 6
Discount depth by price level.

Price	Discount	log(explanation duration)	log(cart availability)
Low	None	−0.219*** (0.034) n = 85,726	0.178*** (0.033)
Low	Deep	−0.548* (0.320) n = 6455	0.073 (0.142)
High	None	−0.832*** (0.072) n = 80,186	0.694*** (0.062)
High	Deep	−2.417*** (0.500) n = 5120	1.188** (0.463)

Notes: Cluster-robust standard errors (session level) in parentheses. ***$p < .01$, **$p < .05$, *$p < .10$. Price is split at the median; deep discount is the upper half of the positive-discount distribution. Each cell is a separate estimation of equation (1) with the full control set and both sets of fixed effects. The four cells sum to 177,487 rather than 188,696: the 11,209 shallow-discount listings are excluded so that the comparison contrasts the two extremes of the discount distribution.

Table 7
Cart availability duration under common delisting.

	All sessions	Common-delist sessions
log(explanation duration)	−0.549*** (0.035)	−0.467*** (0.076)
log(cart availability)	0.448*** (0.031)	0.478*** (0.165)
log(airing number)	0.231*** (0.040)	0.396*** (0.121)
Observations	188,696	39,102
Sessions	12,558	5568

Notes: Cluster-robust standard errors (session level) in parentheses. ***$p < .01$, **$p < .05$, *$p < .10$. Common-delist sessions are those in which every product records an identical removal time. Both columns include the full control set, session fixed effects and streamer–product fixed effects. Estimating the same restriction on the full panel with explanation duration omitted gives an availability coefficient of 0.397 (0.093) on 189,276 listings.

Table 8
Influencer size by category uncertainty.

Follower count	Category	log(explanation duration)	log(cart availability)	Obs.
10–13 million	Low uncertainty	−0.348*** (0.081)	0.211** (0.094)	25,443
	Mid	−0.258*** (0.091)	0.340*** (0.091)	18,741
	High uncertainty	−1.517*** (0.152)	1.240*** (0.123)	18,009
13–20 million	Low uncertainty	−0.437*** (0.137)	0.539*** (0.105)	17,496
	Mid	−0.958*** (0.127)	0.558*** (0.109)	22,636
	High uncertainty	−1.738*** (0.160)	1.258*** (0.152)	21,395
Over 20 million	Low uncertainty	−0.019 (0.018)	0.010* (0.006)	18,379
	Mid	−0.016** (0.006)	0.026** (0.011)	19,935
	High uncertainty	−0.015 (0.013)	0.126** (0.056)	21,836

Notes: Cluster-robust standard errors (session level) in parentheses. ***$p < .01$, **$p < .05$, *$p < .10$. Category return propensity, the proxy for fit uncertainty, is the tercile of the leave-streamer-out category baseline return rate. Each row is a separate estimation of equation (1) with the full control set and both sets of fixed effects. The nine cells sum to 183,870 rather than 188,696 because the category measure is undefined for 2.6 percent of the subsample, as in Panel B of Table 5.

condition. Inspection of the underlying data suggests a more mundane and more consequential explanation. The mean return rate recorded for this tier is 0.20 percent, against 3.22 and 6.67 percent for the other two, and the great majority of its streamers have a mean recorded return rate below one half of one percent. The dependent variable is, for these accounts, close to a constant. Two candidate explanations can be dismissed: it is not a power problem, since the tier contributes 60,150 listings, and it is not a composition effect, since this tier carries the highest share of high-uncertainty merchandise of the three at 36.3 percent against 29.0 and 34.8. We therefore treat the result as a limitation of the vendor's return-rate coverage for a subset of very large accounts rather than as a finding about their commercial behaviour, and we verify that our estimates do not depend on them.

The estimates in Table 9 do not merely survive the exclusion; they roughly double. Dropping the streamers whose recorded returns are near zero moves the explanation coefficient from −0.549 to −1.066 and the availability coefficient from 0.448 to 0.814, while the opposition of signs and the near-equality of magnitudes are preserved throughout. The pooled estimates in Table 2 are smaller in absolute magnitude than those obtained after this exclusion. This comparison shows sensitivity to sample composition; it does not establish that the pooled estimates are statistical lower bounds or identify the reason for near-zero recorded returns. We report the pooled estimates as the baseline and retain the exclusion results as a sensitivity analysis.

Table 9
Excluding streamers with near-zero recorded returns.

Sample restriction	Streamers	log(explanation duration)	log(cart availability)	Obs.
All streamers	160	−0.549*** (0.035)	0.448*** (0.031)	188,696
Drop mean return ≤ 0.1%	86	−0.930*** (0.057)	0.675*** (0.049)	114,961
Drop mean return ≤ 0.5%	79	−1.041*** (0.063)	0.731*** (0.052)	107,049
Drop mean return ≤ 1.0%	69	−1.066*** (0.065)	0.814*** (0.059)	96,688

Notes: Cluster-robust standard errors (session level) in parentheses. ***$p < .01$, **$p < .05$, *$p < .10$. Each row re-estimates the preferred specification after dropping streamers whose mean recorded return rate falls below the stated threshold.

4.5. Allocation

The two durations differ in a way that matters for how a seller should spend a session. Seller explanation duration is associated with favourable movement on both margins at once, while within a session it uses airtime that could be devoted to other products. Cart availability duration is associated with higher sales and lower demand quality, so the trade-off it implies depends on what a return costs the seller. Table 10 estimates both associations separately in each cell of a price-by-category grid and expresses them as a break-even benchmark.

Two conclusions follow, and the second is a caution rather than a prescription. Seller explanation duration is associated with movement in the favourable direction on both margins in every cell — higher sales and lower returns — whether the product is cheap or expensive, search-like or experience-like, and its coefficient on returns grows nearly sevenfold from the least to the most uncertain cell. There is no configuration in these data in which longer explanation is associated with worse outcomes on either margin. Cart availability duration is a trade-off in every cell, and the trade-off tightens monotonically with uncertainty — the break-even cost falls from 45.8 unit margins for cheap search goods to 8.5 for expensive experience goods — but the calculation also suggests that it rarely binds. Read at face value, and only under the assumption that these associations would survive a deliberate intervention, extending shelf time would remain profitable unless handling a return cost more than roughly eight and a half times the unit margin in the least forgiving quadrant, and more than forty-five times in the most forgiving.

One further specification shows that the interaction result is sensitive to the sample restriction. If seller explanation duration works by resolving fit uncertainty before purchase, a viewer who heard the explanation may be better informed than one who did not; purchasing from the cart after the pitch does not itself establish that distinction, but under this interpretation, the positive return-rate association with cart availability could be smaller at longer explanation durations. Interacting the two durations on the estimation sample produces a negative estimate: a coefficient of −0.112 (s.e. 0.018). But the same interaction estimated on the full panel, with the pitched indicator and its own interaction with shelf duration included, is 0.005 (s.e. 0.015), and the implied marginal effect of cart availability duration is flat across the entire range of explanation duration. The interaction is not robust to expanding the sample, so we do not interpret it as evidence of substitution. We report it because it is a prediction of the screening interpretation that is not robustly supported, and because the discrepancy between the two samples is itself informative about how much weight the conditioning can bear.

The honest reading is therefore not that shelf time is a trap. It is that the return cost associated with shelf time is real, is concentrated in an identifiable quadrant, and is nonetheless usually smaller in value than the sales that accompany it. Of the two margins, the estimates make the allocation of the pitch the more promising, not the length of the shelf stay. Seller explanation duration is the only duration measure in these data associated with favourable movement on both outcomes at once, and that association is strongest precisely where the availability association is most adverse. The pattern suggests a possible allocation principle rather than an operational rule: if the associations reflect anything a seller can act on, the expensive, hard-to-judge merchandise in the lower-right cell of Table 10 is where a session's scarce minutes would be worth most. That is a hypothesis for a field test rather than a conclusion of this paper.

5. Robustness checks

The paper rests on a contrast between two coefficients of opposite sign, and a contrast of that kind is only as credible as the specification that produces it. Four choices made in estimating equation (1) could in principle be carrying it rather than the data. The fixed-effect structure

Table 10
Allocation matrix and the break-even cost of a return.

Price	Category	Explanation → returns	Explanation → sales	Availability → returns	Availability → sales
Low	Search-like	−0.173*** (0.032)	0.558*** (0.007)	0.137*** (0.040)	0.264*** (0.010)
Low	Experience-like	−0.299*** (0.088)	0.574*** (0.015)	0.276*** (0.056)	0.162*** (0.012)
High	Search-like	−0.549*** (0.118)	0.596*** (0.011)	0.431*** (0.113)	0.258*** (0.013)
High	Experience-like	−1.170*** (0.083)	0.686*** (0.007)	0.977*** (0.075)	0.185*** (0.009)

Price	Category	Observations	Baseline return rate	Break-even cost c* (× unit margin)
Low	Search-like	61,726	1.61%	45.8
Low	Experience-like	33,326	2.46%	23.1
High	Search-like	30,210	4.40%	15.5
High	Experience-like	58,608	5.26%	8.5
Estimation sample		188,696	3.43%	16.9

Notes: Each cell of the upper panel is a separate estimation of equation (1) on the indicated subsample, with the full control set and both sets of fixed effects; the four cells sum to 183,870 because the category measure is undefined for 2.6 percent of the subsample. Price is split at its median, 59.9 RMB; category is split at the median of the leave-streamer-out category baseline return rate, 2.13 percent, with the lower half labelled search-like and the upper half experience-like, as set out in Section 4.2. The break-even cost is a benchmark, not a policy guarantee: it asks how large return-handling costs would have to be before the sales gain associated with longer shelf time is offset, treating the estimated associations as if a manager could act on them. It solves b_sales·[(1−r) − c·r] = b_ret·(1+c) with the unit margin normalised to one, where b_sales is the availability-duration coefficient on log sales and b_ret is the availability-duration coefficient on the return rate expressed as a PROPORTION, that is the tabulated percentage-point figure divided by one hundred. For the estimation sample this is 0.209·[(1−0.0343) − c·0.0343] = 0.00448·(1+c), which solves to c* = 16.9. Substituting the percentage-point figure directly would yield −0.54, reversing the sign as well as the magnitude.

decides which listings are compared with which. The dependent variable is a ratio, winsorised at the 99th percentile, whose denominator varies by four orders of magnitude and which enters unweighted, so that a listing selling five units counts for as much as one selling fifty thousand. The specification imposes a linear relationship between log(1 + duration) and the return rate. And the level at which standard errors are clustered governs not the estimates themselves but whether they can be told apart from zero.

This section varies each of the four in turn and reports where the estimates move as well as where they do not, because what such an exercise is good for is establishing the boundaries of a result rather than accumulating reassurance about it. It then uses two falsification exercises to assess features of the estimated associations: one breaks the match between explanation time and the product that received it, and the other asks whether a later duration predicts an earlier outcome; neither exercise distinguishes screening from other mechanisms. The reference point throughout is Table 2, column (2): an explanation-duration coefficient of −0.549 and an availability-duration coefficient of 0.448.

5.1. Specification, measurement and inference

Table 11 takes the first three of those choices in turn, one panel each. Panel A varies the fixed-effect structure, moving from the baseline pairing of a session effect with a streamer-product effect through both coarser and finer alternatives, so that the set of listings against which a given listing is compared changes while nothing else about the specification does. Panel B varies the dependent variable and the weighting, since a winsorised ratio estimated unweighted is a choice rather than a necessity, and one that a reader is entitled to see relaxed. Panel C varies the level at which observations are allowed to be correlated; the point estimates there are identical to the baseline by construction, so only the standard errors move.

Panel A. Both coefficients keep their sign, their order of magnitude and their significance in all six structures. Dropping the session effect raises the availability coefficient from 0.448 to 0.661, which is what one expects if broadcasts that keep products on the shelf longer also differ in traffic and promotional intensity; the baseline is therefore the conservative choice. Dropping the streamer-product effect instead compresses both coefficients toward each other, to −0.317 and 0.314: without holding the merchandise constant, part of what the duration variables pick up is which products get pitched. Coarsening the pair effect to a product effect changes almost nothing.

Panel B. The availability coefficient sits between 0.238 and 0.456 across every definition that leaves the dependent variable on its original scale and is significant at the one percent level in all of them; removing the winsorisation, the check most likely to expose a result driven by a few extreme listings, leaves it at 0.456. The explanation coefficient is robust in sign and significance but not in magnitude: it ranges from −0.146 to −0.635, strengthening without winsorisation and weakening both when the upper tail is compressed further and among the highest-volume listings. We read this as an association operating disproportionately through the avoidance of very poor outcomes rather than through a uniform shift of the distribution, and we do not treat the pooled explanation estimate as a constant that applies equally across the sales distribution. One row belongs with the others only as a warning: replacing the rate with an indicator for any return at all makes both coefficients positive, at 4.43 and 6.69, because both durations are associated with higher units sold and the probability that at least one unit comes back rises with the number shipped. That is the extensive margin of returning, which is not the quantity this paper is about.

Panel C. Clustering on the product or the streamer-product pair raises the explanation standard error by about 30 percent and the availability one by 3 percent; two-way clustering on session and product raises them by 40 and 17 percent respectively. The binding case is clustering on the streamer, where there are only 160 clusters: both coefficients remain significant at the one percent level, with *t*-statistics of 2.73 and 5.10, but the margin on the explanation coefficient is thin. Clustering on the streamer produces the largest explanation-coefficient standard error among the clustering schemes reported here.

5.2. Functional form

Both durations enter as log(1 + minutes), so a given proportional change in one plus duration corresponds to the same predicted difference in the return rate. Doubling duration from d to 2d instead changes the transformed regressor by log[(1 + 2d)/(1 + d)]; this depends on the starting duration and approaches log(2) only as duration becomes large. Nothing in the framework requires that, and the two variables have no reason to share a shape merely because they share a transformation. Table 12 therefore replaces both logarithms with quintile indicators, estimated jointly in a single regression so that the comparison between them is held to the same listings, and reports beside each indicator the median duration of the quintile it stands for.

The two variables do not have the same shape, and the logarithm suits one better than the other. For seller explanation duration the coefficients deepen across the distribution and the largest step is at the top, from −0.270 in the fourth quintile to −0.933 in the fifth, whose median

Table 11
Robustness of the two duration coefficients.

	log(explanation duration)	log(cart availability)	Obs.
Panel A. Fixed-effect structure			
Session + streamer × product [baseline]	−0.549*** (0.035)	0.448*** (0.031)	188,696
Session + product	−0.531*** (0.034)	0.429*** (0.031)	188,696
Session + streamer × product + category	−0.548*** (0.035)	0.445*** (0.031)	188,696
Streamer × product only	−0.571*** (0.034)	0.661*** (0.032)	188,696
Session only	−0.317*** (0.025)	0.314*** (0.029)	188,696
Product only	−0.534*** (0.033)	0.665*** (0.031)	188,696
Panel B. Dependent variable, weighting and sample			
Winsorised at 99th percentile [baseline]	−0.549*** (0.035)	0.448*** (0.031)	188,696
No winsorisation	−0.635*** (0.042)	0.456*** (0.035)	188,696
Winsorised at 95th percentile	−0.176*** (0.017)	0.357*** (0.019)	188,696
log(1 + return rate)	−0.014*** (0.003)	0.081*** (0.003)	188,696
Any return (indicator × 100)	4.43*** (0.15)	6.69*** (0.18)	188,696
Weighted by units sold	−0.344*** (0.081)	0.238*** (0.067)	177,253
Weighted by √(units sold)	−0.507*** (0.031)	0.393*** (0.033)	177,253
Listings selling ≥ 100 units	−0.562*** (0.038)	0.449*** (0.034)	120,878
Listings selling ≥ 1000 units	−0.146*** (0.037)	0.383*** (0.048)	52,258
Panel C. Clustering of the standard errors			
Session [baseline]	−0.549*** (0.035)	0.448*** (0.031)	12,558
Streamer	−0.549*** (0.201)	0.448*** (0.088)	160
Streamer × product	−0.549*** (0.045)	0.448*** (0.032)	69,552
Product	−0.549*** (0.045)	0.448*** (0.032)	67,117
Two-way: session and product	−0.549*** (0.049)	0.448*** (0.036)	12,558 /67,117
Two-way: session and streamer	−0.549*** (0.201)	0.448*** (0.088)	12,558 /160

Notes: Cluster-robust standard errors in parentheses; session-level except in Panel C. ***$p < .01$, **$p < .05$, *$p < .10$. Unless stated otherwise every row is the specification of Table 2, column (2), with the full control set, the airing number and both sets of fixed effects. Coefficients in Panel B are not comparable in level to the others in the two rows where the dependent variable is rescaled. Weighted rows drop listings recording no sales. The final column reports observations in Panels A and B and the number of clusters in Panel C. Two-way clustering follows Cameron et al. (2011).

pitch is 24.4 min; a logarithm is a reasonable if slightly conservative summary. For cart availability duration the coefficient jumps to 0.982 in the second quintile and is then flat, at 1.189, 1.014 and 1.029. The association is concentrated in the step out of the shortest quintile — whose median is 8.4 min — rather than in duration beyond it, so the availability coefficient should be read as the average slope of a step rather than of a gradient. This affects no sign or comparison in Section 4, all of which use the same functional form on both sides, but it refines what the coefficient means. Entering the durations in raw minutes gives −0.00245*** (0.00032) and 0.00034*** (0.00010); as square roots, −0.195*** (0.013) and 0.057*** (0.007). Every form preserves the opposition of signs and the significance of both terms.

5.3. Placebo tests, and what these checks settle

Two falsification exercises assess the product-specific association and possible persistent allocation patterns, rather than identifying the screening mechanism. The first reassigns explanation time at random: within each session we permute the recorded durations across that session's products, leaving everything else untouched, and re-estimate. The permuted variable keeps its marginal distribution and its relationship to session characteristics; what it loses is the match between a pitch and the product that received it. Across 40 replications the placebo explanation coefficient has a mean of 0.0056 and a standard deviation of 0.0200, with a range from −0.0311 to 0.0610. The estimate of −0.549 lies 28 placebo standard deviations away, and the most negative placebo draw is one eighteenth of its size. The availability coefficient in the placebo runs settles at 0.319, which is the figure row (d) of Table 3 reports for the specification that omits explanation time — as it should be, since a randomly permuted regressor carries no information to correct the omitted-variable bias.

The second, reported in Table 13, asks whether a decision taken later can predict an outcome recorded earlier. Shelf duration at the next airing is a clean null, −0.011 (0.031), which matters because cart availability duration is the coefficient most exposed to the streamer's discretion. Explanation time at the next airing is small but not zero, −0.082** (0.034), against a contemporaneous coefficient of −0.537 that is roughly six and a half times larger. The smaller coefficient indicates a limitation of the identification strategy and may reflect persistent allocation decisions; it does not resolve which mechanism underlies the contemporaneous association. A streamer who allocates airtime by expected performance will spend more time, airing after airing, on products that sell well, and products that sell well may carry

Table 13
Placebo: Durations at the Pair's next airing.

	Coefficient	t
log(explanation duration), current airing	−0.537*** (0.040)	−13.45
log(cart availability), current airing	0.406*** (0.037)	11.06
log(explanation duration), next airing	−0.082** (0.034)	−2.41
log(cart availability), next airing	−0.011 (0.031)	−0.37

Notes: Cluster-robust standard errors (session level) in parentheses. ***$p < .01$, **$p < .05$, *$p < .10$. The sample is the 119,144 listings whose streamer-product pair airs again later in the panel, across 12,027 sessions. Dependent variable is the return rate of the current listing; both sets of fixed effects and the full control set are included.

Table 12
Quintile indicators in place of the logarithmic form.

Quintile	Median duration	log(explanation duration)	Median duration	log(cart availability)
Q1 (shortest)	0.9 min	reference	8.4 min	reference
Q2	2.9 min	−0.031 (0.067)	64.7 min	0.982*** (0.081)
Q3	5.0 min	−0.249*** (0.072)	136.4 min	1.189*** (0.097)
Q4	9.0 min	−0.270*** (0.078)	258.2 min	1.014*** (0.133)
Q5	24.4 min	−0.933*** (0.092)	824.2 min	1.029*** (0.182)

Notes: Cluster-robust standard errors (session level) in parentheses. ***$p < .01$, **$p < .05$, *$p < .10$. A single regression contains both sets of indicators, the airing number, the full control set and both sets of fixed effects; $n = 188{,}696$. The omitted category is the shortest quintile of each variable. Median durations are the medians of the untransformed minutes within each quintile.

lower return rates for reasons unconnected to the pitch — better merchandise, a better-matched audience, a seller with a stronger after-sales record. The pair fixed effect absorbs what is constant in that allocation but not what drifts with it, and the coefficient on a later decision is the trace of that drift. The ratio of the two coefficients conveys the scale of the contamination without bounding it: a partial coefficient on a later decision measures how much of the allocation persists, not a ceiling on the bias the persistence induces.

Taken together with the design-specific checks of Sections 4.1.1, 4.3 and 4.4 — the alternative handlings of unrecorded explanation time, the common-delisting subsample and the exclusion of streamers with near-zero recorded returns — and with the within-category and within-group comparisons of Section 5.4, which bear on the gradient rather than on the levels — the availability-duration result remains positive and statistically significant in the reported full-sample robustness specifications, and two qualifications attach to its magnitude rather than its existence: Table 12 shows the association is concentrated in the step out of the shortest quintile, and weighting by units sold halves it, from 0.448 to 0.238. The explanation-duration result is robust in sign and significance but not in magnitude, is weaker among the highest-volume listings, rests on the thinnest margin when standard errors are clustered on the streamer, and is the coefficient the second placebo touches. The opposite signs persist across the reported full-sample robustness specifications with return rate as the dependent variable. Subgroup significance and uncertainty gradients are less uniform, as reported in Sections 4.4 and 5.4.

Two limitations no robustness exercise can address. The panel carries no order-level or buyer-level information, so returns are observed as a listing-level rate: we cannot separate a buyer who returned one item from one who returned several, nor observe when in the return window the decision was taken. And nothing here converts an association into a causal effect. The fixed effects, the common-delisting subsample and the placebo tests each narrow the set of explanations that could generate the pattern; they do not close it.

5.4. *Price position within the category, and replication within broad product groups*

The two proxies of Section 4.2 are measured across the whole panel: a listing counts as expensive if it is expensive relative to every other listing, and a category counts as uncertain if its buyers return more than the buyers of other categories. Each measure therefore mixes the position of a product within its own category with the kind of category it belongs to, and the two components need not carry the same information. Price may carry brand tier, return convenience and seller service along with the stakes of a mismatch; the category rate may carry return policy and logistics along with fit. Two exercises separate the within-category component from the between-category one. Category fixed effects are not the instrument, because the streamer–product effect already nests the category — a product belongs to one — so the separation is made instead by measuring price position within the fine category, and by repeating the analysis inside broad product groups.

Table 14 redefines the price quartiles within each fine category, retaining the 735 categories with at least forty listings in the estimation sample, which keeps 164,024 listings and drops 13.1 percent. Panel A reports the result and Panel B repeats the sample-wide quartiles of Table 5 for comparison. The gradient survives and is in one respect cleaner: both coefficients are monotone across the four quartiles, where the sample-wide availability series was not. It is also much flatter. The explanation coefficient runs from −0.427 to −0.654, a ratio of 1.5 between the priciest and the cheapest quartile against 10.2 in Table 5; the availability coefficient runs from 0.254 to 0.700, a ratio of 2.8 against 6.0. Most of the spread that Table 5 attributes to price is therefore between-category composition — expensive categories are different kinds of goods — rather than price position within a category.

Panel C puts the same question in a single equation, interacting each duration with the listing's price percentile within its category, so that the main effects are evaluated at the cheapest listing in the category and the interactions measure how the coefficients change across its price range. The two sides differ. The availability interaction is 0.674 with a standard error of 0.089: the association between shelf time and returns rises steeply with price position within the category, and the coefficient at the cheapest listing, 0.145, is about a fifth of the coefficient at the top. The explanation interaction is −0.154 with a standard error of 0.116 and cannot be distinguished from zero. The monotone explanation series in Panel A is real but small relative to its standard errors, and the pooled test does not reject a flat explanation gradient within category. Within a category, then, the positive return-rate association with cart availability scales with what the buyer stands to lose; the benefit of explanation does not detectably scale in the same way, and its dependence on price in Table 5 is largely a dependence on the kind of category.

Table 14
Price position within the fine category.

	log(explanation duration)	log(cart availability)	Observations
Panel A. Price quartile within the fine category			
Q1 (cheapest in category)	−0.427***	0.254***	41,269
	(0.065)	(0.046)	
Q2	−0.561***	0.341***	40,921
	(0.080)	(0.067)	
Q3	−0.565***	0.516***	40,731
	(0.082)	(0.074)	
Q4 (priciest in category)	−0.654***	0.700***	41,101
	(0.087)	(0.087)	
Panel B. Sample-wide price quartile (Table 5, Panel A, for comparison)			
Q1 (cheapest)	−0.137***	0.200***	50,052
	(0.037)	(0.042)	
Q2	−0.295***	0.177***	46,945
	(0.060)	(0.054)	
Q3	−0.480***	0.407***	44,821
	(0.070)	(0.066)	
Q4 (priciest)	−1.391***	1.202***	46,878
	(0.106)	(0.101)	
Panel C. Single equation with within-category price percentile			
Duration, evaluated at the cheapest listing in the category	−0.499***	0.145***	164,022
	(0.065)	(0.048)	
Duration × price percentile within category	−0.154 (0.116)	0.674***	
		(0.089)	

Notes: Cluster-robust standard errors (session level) in parentheses. ***p < .01, **p < .05, *p < .10. Panels A and C use the 735 fine categories with at least 40 listings in the estimation sample (164,024 listings; 164,022 in Panel C after two listings drop for a missing percentile). Panel A ranks each listing's price among listings of the same fine category. Panel C interacts each duration with the listing's price percentile within its category, scaled to the unit interval, so that the first row is the coefficient at the cheapest listing in a category and the second row is the change across the category's full price range; the percentile enters as a main effect as well (−1.054, SE 0.639). Every estimation includes the airing number, the full control set, session fixed effects and streamer–product fixed effects.

Table 15 repeats the exercise inside five broad product groups formed by keyword rules on the vendor's category labels, together covering two thirds of the estimation sample: apparel and footwear, bags and accessories, beauty and personal care, food and beverage, and standardised goods — daily consumables, household paper, top-up cards and packaged staples whose fit is known before purchase. Panel A estimates equation (1) within each group. The two coefficients carry opposite signs in every group, and the availability coefficient is ordered across groups as fit uncertainty would order them: 1.056 in apparel, 0.862 in bags, 0.777 in beauty, 0.160 in food and 0.115 in standardised goods, where it is the only one of the five not significant at the five percent level. The explanation coefficient is not ordered the same way — food and beverage, at −0.175, is smaller in magnitude than standardised goods, at −0.303 — so the between-group ordering is a property of the

Table 15
Replication within broad product groups.

	log(explanation duration)	log(cart availability)	Observations
Panel A. Equation (1) **within each group**			
Apparel & footwear	−2.380***	1.056***	26,659
	(0.159)	(0.116)	
Bags & accessories	−1.062***	0.862***	7611
	(0.305)	(0.160)	
Beauty & personal care	−0.345***	0.777***	24,361
	(0.071)	(0.098)	
Food & beverage	−0.175***	0.160***	49,777
	(0.041)	(0.046)	
Standardised goods	−0.303** (0.122)	0.115* (0.065)	17,207
Panel B. Split at the group median of price			
Apparel & footwear, below median price	−1.271***	0.252***	13,738
	(0.285)	(0.094)	
Apparel & footwear, above median price	−2.852***	1.992***	12,921
	(0.194)	(0.223)	
Bags & accessories, below median price	−1.115***	0.465**	3826
	(0.336)	(0.187)	
Bags & accessories, above median price	−1.251** (0.491)	1.222***	3785
		(0.260)	
Beauty & personal care, below median price	−0.153** (0.068)	0.470***	12,733
		(0.113)	
Beauty & personal care, above median price	−0.528***	1.121***	11,628
	(0.144)	(0.221)	
Food & beverage, below median price	−0.068* (0.037)	0.045 (0.047)	25,239
Food & beverage, above median price	−0.328***	0.222***	24,538
	(0.089)	(0.079)	
Standardised goods, below median price	−0.316* (0.186)	0.158**	9198
		(0.068)	
Standardised goods, above median price	−0.380** (0.173)	0.025 (0.110)	8009
Panel C. Split at the group median of the category baseline return rate			
Apparel & footwear, low baseline	−1.974***	0.796***	13,218
	(0.213)	(0.153)	
Apparel & footwear, high baseline	−2.474***	1.074***	13,204
	(0.210)	(0.164)	
Bags & accessories, low baseline	−1.279***	1.120***	3780
	(0.438)	(0.346)	
Bags & accessories, high baseline	−0.603* (0.332)	0.628***	3770
		(0.185)	
Beauty & personal care, low baseline	−0.340***	0.841***	12,081
	(0.115)	(0.166)	
Beauty & personal care, high baseline	−0.403***	0.852***	12,080
	(0.099)	(0.142)	
Food & beverage, low baseline	−0.172***	0.240***	24,792
	(0.065)	(0.079)	
Food & beverage, high baseline	−0.175***	0.108* (0.056)	24,778
	(0.048)		
Standardised goods, low baseline	−0.023 (0.088)	0.052 (0.100)	8618
Standardised goods, high baseline	−0.739** (0.311)	0.273***	8565
		(0.096)	

Notes: Cluster-robust standard errors (session level) in parentheses. ***p < .01, **p < .05, *p < .10. Groups are formed by keyword rules applied to the vendor's fine category labels; they contain 227, 68, 135, 524 and 107 fine categories respectively and together cover 125,616 of the 188,696 listings, the remainder (33.4 percent) falling in categories — books, homeware, tea, bundles — that none of the five rules assigns. Panel B splits each group at its own median listed price; Panel C at its own median of the leave-streamer-out category baseline return rate, which is undefined for 4.8 percent of listings. Every estimation includes the airing number, the full control set, session fixed effects and streamer–product fixed effects.

availability coefficient rather than of both.

Panel B splits each group at its own median price. In four of the five the gradient runs the predicted way on both coefficients, most steeply in apparel, where the explanation coefficient moves from −1.271 to −2.852 and the availability from 0.252 to 1.992, and in food and beverage the low-price half is the predicted null, with neither coefficient significant. Two entries do not fit. In bags and accessories the availability coefficient nearly triples across the split while the explanation coefficient barely moves. In standardised goods the availability gradient runs backwards, from 0.158 to 0.025, though both figures are close to zero and neither the level nor the direction is well determined in a group with little to screen. One further feature should be stated. Beauty and personal care is the one group in which the availability coefficient exceeds the explanation one in absolute value, 0.777 against 0.345, the reverse of the pooled result; we have no account of this beyond the observation that fit to skin type may be a form of uncertainty a pitch resolves less well than it resolves size.

Panel C splits each group at the median of the leave-streamer-out category baseline return rate. Held within a broad group, the category proxy largely stops sorting listings the way it does across the whole sample: the predicted gradient appears on both coefficients in apparel and in standardised goods, is flat in beauty, and runs backwards on the availability side in food and on both sides in bags. This is consistent with the account of the measure in Section 4.2. Its information is between-category — the difference between shoes and tissues — and once that difference is held within a group, what remains is the realised return propensity of neighbouring categories, in which policy and logistics are a larger share.

What the two tables leave standing is narrower than Table 5 and more clearly located. The availability association increases with within-category price position, but the corresponding explanation gradient is not statistically distinguishable from zero. Across broad product groups, the availability coefficients generally follow the expected ordering, whereas the explanation coefficients do not. Within-group comparisons vary across groups and across the price and category-based proxies, including flat or reversed gradients; the evidence therefore does not uniformly support the predicted joint pattern. Section 6.1 discusses how this narrower pattern limits the empirical support for the framework's conditional predictions.

6. Conclusion

Livestream selling combines seller explanation with continued cart availability. Explanation may convey information alongside persuasion, endorsement, and interaction; availability does not establish what viewers saw or learned. Both durations are positively associated with sales, which alone would not reveal their contrasting return-rate associations. On the return rate they diverge in opposite directions by comparable amounts, and the divergence widens as the scope for uncertainty about fit increases. One interpretation is that these activities relate differently to buyer composition, but neither duration identifies purchasers' information or exposure to an explanation.

6.1. Theoretical implications

Prior research has distinguished forms of livestream communication and documented that stronger purchasing outcomes can coexist with poorer postpurchase outcomes. Our contribution is the joint comparison of seller explanation duration and cart availability duration on sales and return rates within the same empirical specification. We examine separately recorded explanation and cart availability durations, using reach and screening as possible underlying processes rather than as measured constructs. The distinction matters because the two durations need not have the same associations with postpurchase outcomes, even though neither isolates the information consumers receive. We find longer cart availability associated with a higher return rate, by an amount comparable to the negative association with explanation duration, but the opposite signs do not identify their underlying mechanisms. The pattern is consistent with a buyer-screening interpretation, but does not uniquely support it. Differences in information acquisition, persuasion, or telepresence could also contribute, and the listing-level data neither identify their relative roles nor establish the proposed division between screened and unscreened purchasers.

Two consequences follow for other work. The first concerns the

conditional prediction that the two associations diverge with fit uncertainty and become smaller where it is negligible: these patterns can assess consistency with the screening illustration, but cannot distinguish it from other explanations. Our evidence meets the first part across two differently constructed proxies at the level of the whole sample, and the second only partly, since the coefficients approach zero in the cheapest decile of merchandise but remain distinguishable from zero at the bottom of the category-based measure. Section 5.4 narrows the first part further: once price position is measured within the fine category, the availability coefficient still scales with it and the explanation coefficient does not detectably do so. The stricter comparison therefore provides only partial support for the predicted joint pattern: the positive availability association varies with within-category price position, while variation in the negative explanation association occurs mainly across kinds of goods. The second concerns outcome measurement. In our setting, positive sales associations for both durations coexist with opposite return-rate associations. Considering sales and returns together therefore reveals a difference in postpurchase outcomes that sales alone would miss, without implying that prior research has ignored communication heterogeneity.

The contrasting return-rate associations suggest that sales alone provide an incomplete basis for assessing seller communication and product availability. This comparison draws together the distinction between access to alternatives and support for their evaluation, and the role of search in determining when consumers resolve uncertainty about product match (Alba et al., 1997; Jerath & Ren, 2025). Our findings examine these issues through two recorded selling activities: explanation and availability durations are both positively associated with sales, yet have opposite associations with return rates. Evaluating these activities jointly against purchases and returns can therefore reveal differences that their sales associations alone would obscure. Whether similar patterns arise in other retail settings remains an open question, and our data do not establish which attention, consideration, or evaluation processes account for the observed differences.

6.2. Managerial implications

For a streamer, the findings suggest different considerations for allocating explanation time and setting cart availability duration. Explanation is associated with better outcomes on both margins in every segment we examine, while within a broadcast it uses airtime that could be devoted to other products; given a fixed broadcast length, the allocation question is where the minutes do most good, and the estimates point to where uncertainty about fit is greatest. The question is one of division rather than of inclusion: the estimates concern how long to explain products the host has already decided to explain, and say nothing about whether to add a product to that list, since the products a streamer chooses to pitch differ from those left in the cart in ways the design does not separate from the pitch. Shelf time is a trade-off rather than a dominant instrument, but a looser one than the size of its coefficient suggests: taken at face value, the estimates imply that handling a return would have to cost between roughly 8.5 and 45.8 times the unit margin, depending on the quadrant, before extending a product's stay stopped paying for itself.

One qualification narrows the advice and makes it more useable. Shelf duration is not a lever a seller pulls product by product, because carts are cleared in bulk at the close of a broadcast; what a seller controls is the order of the running list and the division of the explanation minutes across it. A merchant placing the same product with many streamers should read the result the same way. Additional streamer placements create purchase opportunities, but the duration measures do not establish their realised reach or the screening consumers experienced. Contracts written on gross merchandise value pay for the first component and leave the cost of the second with the merchant — precisely the asymmetry a quantity-based measure of performance cannot see.

One assumption underlies this advice and should be stated, because it may not hold for long. The role of cart availability may depend on interface design. A listing can convey information, and availability duration does not establish how well viewers could assess fit from it. Platforms are adding tools — augmented-reality try-on, virtual placement of a product in the buyer's own space, interactive simulation of its use — that let a viewer resolve fit without claiming any of the host's attention, and research on augmented reality locates its value to consumers in exactly that reduction of uncertainty (Rauschnabel et al., 2024; Barta et al., 2025). Under a screening interpretation, tools that help buyers assess fit before purchase could narrow the gap between the two associations, especially where fit is difficult to judge. This is a hypothesis for future research, since such tools could also change attention, persuasion, or telepresence; our data do not establish their benefits or the processes through which they would operate.

6.3. Limitations and future research

Three features of the data bound what the paper can claim. The panel carries no order-level or buyer-level identifiers, so a return is a rate attached to a listing rather than an event attached to a person; we cannot follow a buyer across purchases, nor partition buyers by decision style, which is what testing whether an impulse channel operates within a particular segment would require (Section 4.2.1); we therefore cannot ask whether the recorded durations are associated with differences in subsequent customer value — the question our own results make most pressing. The vendor records total explanation minutes but not their timing within the availability window or whether individual purchasers heard the explanation before buying; even precise timing alone would not establish screening, which requires evidence on exposure, fit assessment, and selection into purchase. And it does not disclose the window over which a return is attributed to a listing. Two features of the estimates travel with them: the recorded return rate sits close to zero for a substantial group of streamers, and excluding them roughly doubles both coefficients, to −1.066 and 0.814, indicating sensitivity to this sample restriction rather than establishing a statistical lower bound; and the design identifies associations rather than effects, which an experiment randomising the allocation of explanation minutes across a running list would be needed to settle.

Four extensions follow. Comparing communication and availability in other channels could establish whether similar associations arise beyond this format, without presuming that their underlying processes are the same. Whether merchants who pay on gross merchandise value systematically over-purchase availability without accompanying seller explanation is measurable with data on placement terms. And a setting containing a large volume of genuinely search-like merchandise would test the prediction that both coefficients vanish where fit is knowable before purchase, which our panel can only gesture at. And the framework specifies the conditions under which its central asymmetry should shrink. Where a platform introduces tools that let a buyer establish fit from the cart — augmented-reality try-on is the current case — the availability coefficient should fall toward the explanation one, category by category, in step with how much those tools resolve. A rollout with credible exogenous variation could test the feature's effects, but distinguishing screening from attention or telepresence would additionally require purchaser-level process measures or interventions that separate those mechanisms. What the paper establishes, within those bounds, is narrow and we think durable. Seller explanation duration and cart availability duration have different associations with return rates despite positive associations with sales. Their joint analysis shows why sales alone may miss differences in postpurchase outcomes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Appendix

Table A1
Variable Definitions

Variable	Definition	Coverage
Return rate y	Returned units divided by units sold for that listing, in percentage points, winsorised at the 99th percentile of the full 617,440-listing panel, a cap of 59.05 percentage points.	100%
log(explanation duration)	log(1 + minutes the host spent explaining the product in that session). May include information, persuasion, endorsement, demonstration, and interaction; not a measure of information received.	100%
log(cart availability)	log(1 + minutes between the product being added to and removed from the pinned cart). May overlap with explanation; measures purchase availability, not actual viewing or attention.	100%
log(airing number)	log of the ordinal position of this airing within the streamer–product pair's history, ordered by listing time; the pair's first airing takes the value 1.	100%
log(sales)	log(1 + units sold in that listing).	100%
log(price)	log(1 + the coupon-inclusive price actually payable, in RMB).	100%
Discount depth	1 − (coupon price ÷ list price), clipped to [0, 1]; zero when no list price is recorded or the coupon price is not below it.	100%
Branded listing	One if the vendor records a brand for the product, zero otherwise.	100%
Relative position	Percentile rank of the listing within its session's running order, so that 0 is the first product of the session and 1 the last.	100%
Category baseline return rate	Mean return rate of the listing's category over all OTHER streamers in the panel, so it carries no information about the focal streamer's own outcomes; undefined when no other streamer carries that category.	97.4%
log(product age)	log(1 + days since the vendor-recorded date on which the product was first listed anywhere on the platform, floored at zero; that date can precede the panel window by up to two years, and for 7.3 percent of listings it postdates the listing itself.)	100%
Commission rate	Vendor-recorded commission percentage for the listing.	79.9%

Notes: Coverage is the share of the 188,698 listings in the estimation sample of Section 4 for which the variable is defined; explanation duration is recorded for every listing in that sample by construction, and for 30.6 percent of the full panel. All duration variables enter as log(1 + minutes) rather than log(minutes) so that slots of arbitrarily short duration are retained. The panel distinguishes 3463 product categories, 2971 of which appear in the estimation sample, so the category measure is computed at the level of running shoes rather than footwear. Discount bins used in Section 4 are: none, covering 87.9 percent of listings, and shallow versus deep split at the median depth among discounted listings, 0.334.

Data availability

The data that has been used is confidential.

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